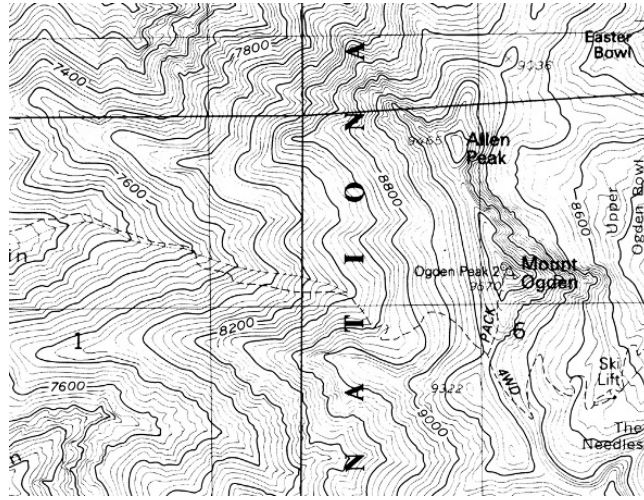




**FIGURE 17.16** Topographical map with elevation contour lines. The shortest distance between two points on the map may not be a straight line. (Courtesy of USGS.)

map when planning a hike up a steep trail. The additional information provided by the map's elevation contour lines must be included in any calculation of the actual hiking distance, which is always greater than the difference in map coordinates; see Fig. 17.16.

The Schwarzschild metric also incorporates time dilation and the gravitational redshift (two aspects of the same effect). If a clock is at rest at the radial coordinate  $r$ , then the proper time  $d\tau$  it records (Eq. 17.17) is related to the time  $dt$  that elapses at an infinite distance by

$$d\tau = \frac{ds}{c} = dt \sqrt{1 - \frac{2GM}{rc^2}}, \quad (17.24)$$

which is, of course, just Eq. (17.13). Since  $d\tau < dt$ , this shows that time passes more slowly closer to the planet.

### The Orbit of a Satellite

Having finally learned how a spherical object of any mass acts on spacetime, telling it how to curve, we are now ready to calculate how curved spacetime acts on a particle, telling it how to move. The rest of this section will be devoted to using general relativity to find the motion of a satellite about the planet. All we need is the rule that it will follow the straightest possible worldline, the worldline with an *extremal* interval.<sup>11</sup>

At this point, you may be fondly recalling the simplicity of Newtonian gravity. According to Newton, the motion of a satellite in a circular orbit around Earth is found by simply equating the centripetal and gravitational accelerations. That is,

$$\frac{v^2}{r} = \frac{GM}{r^2},$$

<sup>11</sup>It is assumed that the satellite's mass  $m$  is small enough that its effect on the surrounding spacetime is negligible.

where  $v$  is the orbital speed. This immediately results in

$$v = \sqrt{\frac{GM}{r}}.$$

Einstein and Newton must agree in the limiting case of weak gravity, so this result must be concealed within the Schwarzschild metric for curved spacetime.<sup>12</sup> It can be found by using the Schwarzschild metric to find the straightest possible worldline for the satellite's circular orbit.

Powerful tools are available for calculating the worldline with the maximum interval between two fixed events. If we employed such an approach, the orbit of the satellite would emerge along with the laws of the conservation of energy, momentum, and angular momentum because they are built into Einstein's field equations. However, we will use a simpler strategy and assume from the beginning that the satellite travels above Earth's equator ( $\theta = 90^\circ$ ) in a circular orbit with a specified angular speed  $\omega = v/r$ . Inserting these choices, along with  $dr = 0$ ,  $d\theta = 0$ , and  $d\phi = \omega dt$ , into the Schwarzschild metric gives

$$(ds)^2 = \left[ \left( c\sqrt{1 - 2GM/rc^2} \right)^2 - r^2\omega^2 \right] dt^2 = \left( c^2 - \frac{2GM}{r} - r^2\omega^2 \right) dt^2.$$

Integrating, the spacetime interval for one orbit is just

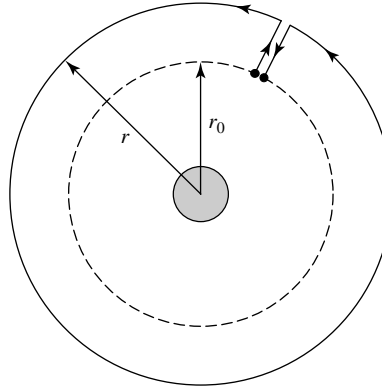
$$\Delta s = \int_0^{2\pi/\omega} \sqrt{c^2 - \frac{2GM}{r} - r^2\omega^2} dt. \quad (17.25)$$

When finding the value of  $r$  for which the interval is an extremum, we must be certain that the endpoints of the satellite's worldline remain fixed. That is, the satellite's orbit must always begin and end at the same position,  $r_0$ , for all of the worldlines. To accommodate orbits of different radii, consider the "orbit" shown in Fig. 17.17. We start the satellite at  $r_0$  and then move it (at nearly the speed of light) radially outward to the radius  $r$  of its actual orbit. At the end of the orbit, the satellite returns just as rapidly to its starting point at  $r_0$ . Fortunately, the quick radial excursions at the beginning and the end of the orbit can be made with negligible contribution to the integral for the spacetime interval. (At almost the speed of light, the contribution is nearly null.) The net effect is a purely circular motion, so Eq. (17.25) can be used to evaluate the interval.

In Eq. (17.25), the limits of integration are constant and the only variable is  $r$ . The value of the radial coordinate  $r$  for the orbit actually followed by the satellite must be the one for which  $\Delta s$  is an extremum. This value may be found by taking the derivative of  $\Delta s$  with respect to  $r$  and setting it equal to zero:

$$\frac{d}{dr}(\Delta s) = \frac{d}{dr} \left( \int_0^{2\pi/\omega} \sqrt{c^2 - \frac{2GM}{r} - r^2\omega^2} dt \right) = 0.$$

<sup>12</sup>To avoid succumbing to Newtonian nostalgia, you should remember that when Einstein and Newton disagree, nature sides with Einstein.



**FIGURE 17.17** The “orbit” of a satellite, showing the radial motions used to keep the endpoints of the satellite’s worldline fixed. The net effect is a circular orbit.

The derivative may be taken inside the integral to obtain

$$\frac{d}{dr} \sqrt{c^2 - \frac{2GM}{r} - r^2 \omega^2} = 0,$$

implying

$$\frac{2GM}{r^2} - 2r\omega^2 = 0.$$

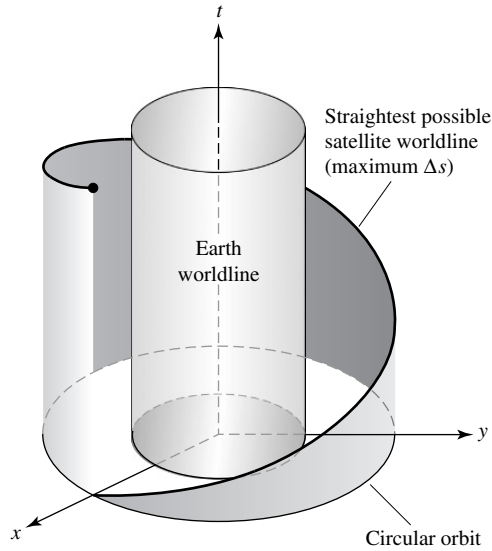
Thus, as promised,

$$v = r\omega = \sqrt{\frac{GM}{r}} \quad (17.26)$$

is the coordinate speed of the satellite for a circular orbit. [By coordinate speed, we simply mean that  $v = r d\phi/dt$  is speed of the satellite measured in the  $(r, \theta, \phi, t)$  coordinate system used by a distant observer.] Figure 17.18 illustrates how this straightest possible worldline through curved spacetime is projected onto the orbital plane, resulting in the satellite’s circular orbit around Earth. In fact, this result is valid even for the very large spacetime curvature encountered around a black hole.

### 17.3 ■ BLACK HOLES

In 1783 John Michell (1724–1793), an English clergyman and amateur astronomer, considered the implications of Newton’s corpuscular theory of light. If light were indeed a stream of particles, then it should be influenced by gravity. In particular, he conjectured that the gravity of a star 500 times larger than the Sun, but with the Sun’s average density, would be sufficiently strong that even light could not escape from it. As you may verify using Eq. (2.17), the escape velocity of Michell’s star would be the speed of light. Naively



**FIGURE 17.18** The straightest possible worldline through curved spacetime and its projection onto the orbital plane of the satellite.

setting the Newtonian formula for the escape velocity equal to  $c$  shows that  $R = 2GM/c^2$  is the radius of a star whose escape velocity equals the speed of light. In terms of the mass of the Sun,  $R = 2.95(M/M_\odot)$  km. Even if this Newtonian derivation were correct, the resulting radius of such a star seemed unrealistically small, and so it held little interest for astronomers until the middle of the twentieth century.

In 1939 American physicists J. Robert Oppenheimer and Hartland Snyder (1913–1962) described the ultimate gravitational collapse of a massive star that had exhausted its sources of nuclear fusion. It was earlier that year that Oppenheimer and Volkoff had calculated the first models of neutron stars (see Section 16.6). We have seen that a neutron star cannot be more massive than about  $3 M_\odot$ .<sup>13</sup> Oppenheimer and Snyder pursued the question of the fate of a degenerate star that might exceed this limit and surrender completely to the force of gravity.

### The Schwarzschild Radius

For the simplest case of a nonrotating star, the answer lies in the Schwarzschild metric, Eq. (17.22):

$$(ds)^2 = \left( c dt \sqrt{1 - 2GM/rc^2} \right)^2 - \left( \frac{dr}{\sqrt{1 - 2GM/rc^2}} \right)^2 - (r d\theta)^2 - (r \sin \theta d\phi)^2.$$

<sup>13</sup>As previously discussed, the upper mass limit of a neutron star is between  $2.2 M_\odot$  and  $2.9 M_\odot$  depending on the amount of rotation; recall Sections 15.4 and 16.6. We will adopt an approximate value of  $3 M_\odot$  for the purposes of this discussion.

When the radial coordinate of the star's surface has collapsed to

$$R_S = 2GM/c^2, \quad (17.27)$$

called the **Schwarzschild radius**, the square roots in the metric go to zero. The resulting behavior of space and time at  $r = R_S$  is remarkable. For example, according to Eq. (17.17), the proper time measured by a clock at the Schwarzschild radius is  $d\tau = 0$ . Time has slowed to a complete stop, as measured from a vantage point that is at rest a great distance away.<sup>14</sup> From this viewpoint, *nothing ever happens at the Schwarzschild radius!*

This behavior is quite curious; does it imply that even light is frozen in time? The speed of light measured by an observer suspended above the collapsed star must always be  $c$ . But from far away, we can determine that light is delayed as it moves through curved spacetime. (Recall the time delay of radio signals from the Viking lander on Mars described in Section 17.1.) The apparent speed of light, the rate at which the spatial coordinates of a photon change, is called the *coordinate speed of light*. Starting with the Schwarzschild metric with  $ds = 0$  for light,

$$0 = \left( c \, dt \sqrt{1 - 2GM/rc^2} \right)^2 - \left( \frac{dr}{\sqrt{1 - 2GM/rc^2}} \right)^2 - (r \, d\theta)^2 - (r \sin \theta \, d\phi)^2,$$

we can calculate the coordinate speed of a vertically traveling photon. Inserting  $d\theta = d\phi = 0$  shows that, in general, the coordinate speed of light in the radial direction is

$$\frac{dr}{dt} = c \left( 1 - \frac{2GM}{rc^2} \right) = c \left( 1 - \frac{R_S}{r} \right). \quad (17.28)$$

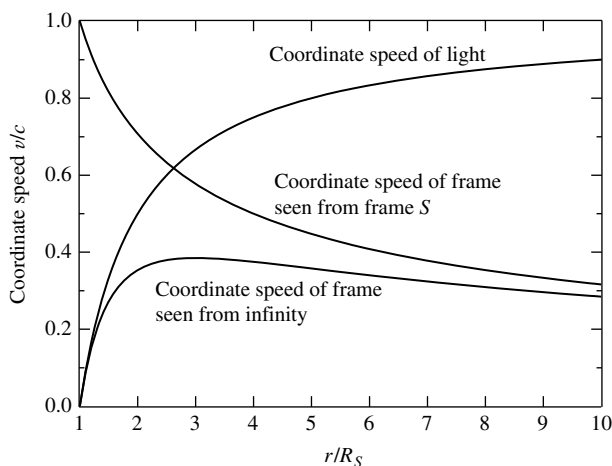
When  $r \gg R_S$ ,  $dr/dt \simeq c$ , as expected in flat spacetime. However, at  $r = R_S$ ,  $dr/dt = 0$  (see Fig. 17.19). Light is indeed frozen in time at the Schwarzschild radius. The spherical surface at  $r = R_S$  acts as a barrier and prevents our receiving any information from within. For this reason, a star that has collapsed down within the Schwarzschild radius is called a **black hole**.<sup>15</sup> It is enclosed by the **event horizon**, the spherical surface at  $r = R_S$ . Note that the event horizon is a mathematical surface and need not coincide with any physical surface.

Although the interior of a black hole, inside the event horizon, is a region that is forever hidden from us on the outside, its properties may still be calculated. A nonrotating black hole has a particularly simple structure. At the center is the **singularity**, a point of zero volume and infinite density where all of the black hole's mass is located. Spacetime is infinitely curved at the singularity.<sup>16</sup> Cloaking the central singularity is the event horizon,

<sup>14</sup>You should recall that the spacetime coordinates  $(r, \theta, \phi, t)$  in the Schwarzschild metric were established for use by an observer at rest at  $r \simeq \infty$ .

<sup>15</sup>The term *black hole* is the 1968 invention of the American theoretical physicist John A. Wheeler.

<sup>16</sup>The black hole's singularity is a real physical entity. It is not a mathematical artifact, as is the *mathematical* singularity exhibited by the Schwarzschild metric at the event horizon (where  $1/\sqrt{1 - 2GM/rc^2} \rightarrow \infty$ ). Choosing another coordinate system would remove the divergence at the event horizon, so that divergence has no physical significance.



**FIGURE 17.19** Coordinate speed of light, and coordinate speeds of a freely falling frame  $S$  seen by an observer at rest at infinity and by an observer in the frame  $S$ . The radial coordinates are in terms of  $R_S$  for a  $10 M_\odot$  black hole having a Schwarzschild radius of  $\approx 30$  km.

so the singularity can never be observed. In fact, there is a hypothesis dubbed the “Law of Cosmic Censorship” that forbids a *naked singularity* from appearing unclothed (without an associated event horizon).

### A Trip into a Black Hole

An object as bizarre as a black hole deserves closer scrutiny. Imagine an attempt to investigate the black hole by starting at a safe distance and reflecting a radio wave from an object at the event horizon. How much time will it take for a radio photon (or any photon) to reach the event horizon from a radial coordinate  $r \gg R_S$  and then return? Since the round trip is symmetric, it is necessary only to find the time for either the journey in or out and then double the answer. It is easiest to integrate the coordinate speed of light in the radial direction, Eq. (17.28), between two arbitrary values of  $r_1$  and  $r_2$  to obtain the general answer,

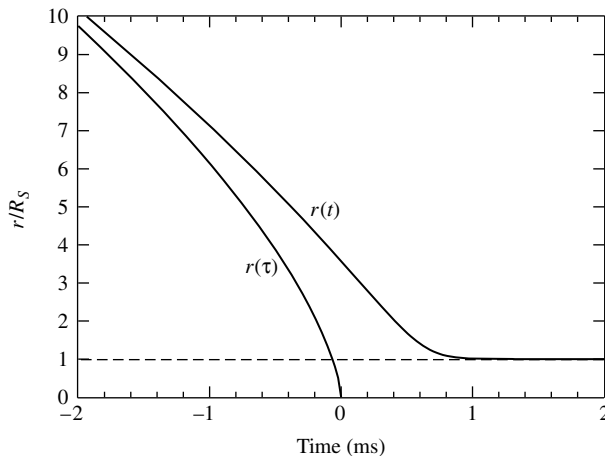
$$\Delta t = \int_{r_1}^{r_2} \frac{dr}{dr/dt} = \int_{r_1}^{r_2} \frac{dr}{c(1 - R_S/r)} = \frac{r_2 - r_1}{c} + \frac{R_S}{c} \ln \left( \frac{r_2 - R_S}{r_1 - R_S} \right),$$

assuming that  $r_1 < r_2$ . Inserting  $r_1 = R_S$  for the photon’s original position, we find that  $\Delta t = \infty$ . Now, since the trip is symmetric, the same result applies if the photon started at  $R_S$ . According to the distant observer, the radio photon will *never* reach the event horizon. Instead, according to gravitational time dilation, the photon’s coordinate velocity will slow down until it finally stops at the event horizon in the infinite future. In fact, any object falling toward the event horizon will suffer the same fate. Seen from the outside, even the surface of the star that collapsed to form the event horizon would be frozen, and so a black hole is in this sense a *frozen star*.

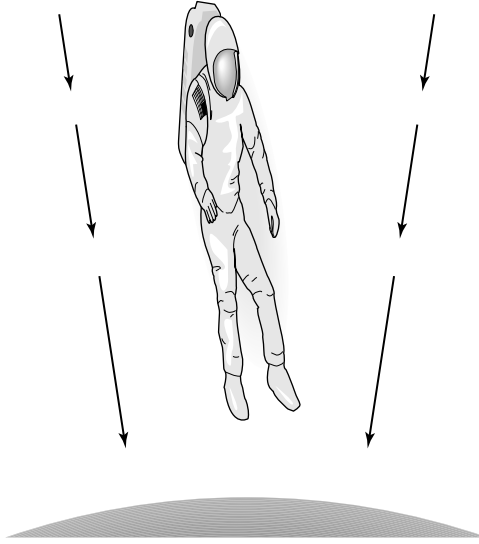
A brave (and *indestructible*) astronomer decides to test this remarkable conclusion. Starting from rest at a great distance, she volunteers to fall freely toward a  $10 M_\odot$  black hole

( $R_S \simeq 30$  km). We remain behind to watch her local inertial frame  $S$  as it falls with coordinate speed  $dr/dt$  all the way to the event horizon. She gradually accelerates as she monitors her watch and shines a monochromatic flashlight back in our direction once every second. As her fall progresses, the light signals arrive farther and farther apart for several reasons: Subsequent signals must travel a longer distance as she accelerates, and her proper time  $\tau$  is running more slowly than our coordinate time  $t$  due to her location (gravitational time dilation) and her motion (special relativity time dilation). Furthermore, the coordinate speed of light becomes slower as she approaches the black hole, so the signals travel back to us more slowly. The frequency of the light waves we receive is also increasingly redshifted. This is caused by both her acceleration away from us and the gravitational redshift. The light becomes dimmer as well, as the rate at which her flashlight emits photons decreases (seen from our vantage point) and the energy per photon ( $hc/\lambda$ ) also declines. Then when she is about  $2R_S$  from the event horizon, the time between her signals begins to increase without limit as the strength of the signals decreases. The light is redshifted and dimmed into invisibility as time dilation brings her coordinate speed to zero (see Figs. 17.19 and 17.20). She is frozen in time, held for eternity like a fly caught in amber. Our successors could watch for millennia while stars were born, evolved, and died without receiving a single photon from her.

How does all of this appear to the brave astronomer, freely falling toward the black hole? Because gravity has been abolished in her local inertial frame, initially she does not notice her approach to the black hole. She monitors her watch (which displays her proper time,  $\tau$ ), and she turns on her flashlight once per second. However, as she draws closer, she begins to feel as though she is being stretched in the radial direction and compressed in the perpendicular directions; see Fig. 17.21. The gravitational pull on her feet (nearer the black hole) is stronger than on her head, and the variation in the direction of gravity



**FIGURE 17.20** Coordinate  $r(t)$  of a freely falling frame  $S$  according to an observer at rest at infinity, and  $r(\tau)$  according to an observer in the frame  $S$ . The radial coordinates are in terms of  $R_S$  for a  $10 M_\odot$  black hole.



**FIGURE 17.21** Tidal forces near a black hole.

from side to side produces a compression that is even more severe. These differential *tidal forces* increase in strength as she falls. In other words, the size of her local inertial frame (where gravity has been abolished) becomes increasingly smaller as the spatial variation in the gravitational acceleration vector,  $\mathbf{g}$ , increases. Were she not indestructible, our astronomer would be torn apart by the tidal force while still several hundred kilometers from the black hole.<sup>17</sup>

In just two milliseconds (proper time), she falls the final few hundred kilometers to the event horizon and crosses it. Her proper time continues normally, and she encounters no frozen stellar surface since it has fallen through long ago.<sup>18</sup> However, once inside the event horizon, her fate is sealed. It is impossible for any particle to be at rest when  $r < R_S$ , as can be seen from the Schwarzschild metric (Eq. 17.22). Using  $dr = d\theta = d\phi = 0$  for an object at rest, the interval is given by

$$(ds)^2 = (c dt)^2 \left( 1 - \frac{R_S}{r} \right) < 0$$

when  $r < R_S$ . This is a *spacelike* interval, which is not permitted for particles. Therefore it is impossible to remain at rest where  $r < R_S$ . Within the event horizon of a nonrotating black hole, all worldlines converge at the singularity. Even photons are pulled in toward the center. This means that the astronomer never has an opportunity to glimpse the singularity because no photons can reach her from there. She can, however, see the light that falls in behind her from events in the outside universe, but she does not see the entire history of

<sup>17</sup>You may recall Problem 16.11 concerning the stretching of an iron cube near a neutron star.

<sup>18</sup>The presence or absence of a frozen stellar surface at the event horizon makes no real difference; the Schwarzschild metric specifies the same spacetime curvature outside.



the universe as it unfolds. Although the elapsed coordinate time in the outside world does become infinite, the light from all of these events does not have time to reach the astronomer. Instead, these events occur in her “elsewhere.” Just  $6.6 \times 10^{-5}$  s of proper time after passing the event horizon, she is inexorably drawn to the singularity.<sup>19</sup>

### Mass Ranges of Black Holes

Black holes appear to exist with a range of masses. **Stellar-mass black holes**, with masses in the range of 3 to  $15 M_{\odot}$  or so, may form directly or indirectly as a consequence of the core-collapse of a sufficiently massive supergiant star, as discussed in Chapter 15. The direct collapse of the core to a black hole may be responsible for the production of collapsar, whereas a delayed collapse of a rapidly rotating neutron star may result in a supranova. It is also possible that a neutron star in a close binary system may gravitationally strip enough mass from its companion that the neutron star’s self-gravity exceeds the ability of the degeneracy pressure to support it, again resulting in a black hole.

**Intermediate-mass black holes** (IMBHs) may exist that range in mass from roughly  $100 M_{\odot}$  to in excess of  $1000 M_{\odot}$  (or perhaps even greater than  $10^4 M_{\odot}$ ). Evidence for them exists in the detection of sources known as **ultraluminous X-ray sources** (ULXs) that have been discovered by satellites such as Chandra and XMM-Newton. It is not entirely clear how these objects might form, although the correlation of IMBHs with the cores of globular clusters and low-mass galaxies suggests that they may develop in these dense stellar environments either by the mergers of stars to form a supermassive star that then core-collapses, or by the merger of stellar-mass black holes.

**Supermassive black holes** (SMBH) are known to exist at the centers of many (and probably most) galaxies. These enormous black holes range in mass from  $10^5 M_{\odot}$  to  $10^9 M_{\odot}$  (our own Milky Way Galaxy has a central black hole of mass  $M = 3.7 \pm 0.2 \times 10^6 M_{\odot}$ ). How these behemoths formed remains an open question. One popular suggestion is that they formed from collisions between galaxies; another is that they formed as an extension of the formation process of IMBHs. Whatever the process, SMBHs appear to be closely linked with some bulk properties of galaxies, implying an important connection between galaxy formation and the formation of SMBHs.

Black holes may have also been manufactured in the earliest instants of the universe. Presumably, these **primordial black holes** would have been formed with a wide range of masses, from  $10^{-8}$  kg to  $10^5 M_{\odot}$ . The only criterion for a black hole is that its entire mass must lie within the Schwarzschild radius, so the Schwarzschild metric is valid at the event horizon.

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**Example 17.3.1.** If Earth could somehow (miraculously) be compressed sufficiently to become a black hole, its radius would only be  $R_S = 2GM_{\oplus}/c^2 = 0.009$  m. Although a primordial black hole could be this size, it is almost impossible to imagine packing Earth’s entire mass into so small a ball.<sup>20</sup>

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<sup>19</sup>A thorough description of the final view of the falling astronomer may be found in Rothman et al. (1985).

<sup>20</sup>You are reminded that the Schwarzschild metric is valid only outside matter. It does not describe the spacetime inside Earth.

### Black Holes Have No Hair!

Whatever the formation processes of black holes, they are certain to be very complicated. For example, the core-collapse of a star is almost certainly not symmetrical. Detailed calculations have demonstrated, however, that any irregularities are radiated away by gravitational waves (see Section 18.6). As a result, once the surface of the collapsing star reaches the event horizon, the exterior spacetime horizon is spherically symmetric and described by the Schwarzschild metric.

Another complication is the fact that all stars rotate, and therefore so will the resulting black hole. Remarkably, however, any black hole can be completely described by just three numbers: its mass, angular momentum, and electric charge.<sup>21</sup> Black holes have no other attributes or adornments, a condition commonly expressed by saying that “a black hole has no hair.”<sup>22</sup>

There is a firm upper limit for a rotating black hole’s angular momentum given by

$$L_{\max} = \frac{GM^2}{c}. \quad (17.29)$$

If the angular momentum of a rotating black hole were to exceed this limit, there would be no event horizon and a naked singularity would appear, in violation of the Law of Cosmic Censorship.

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**Example 17.3.2.** The maximum angular momentum for a solar-mass black hole is

$$L_{\max} = \frac{GM_{\odot}^2}{c} = 8.81 \times 10^{41} \text{ kg m}^2 \text{ s}^{-1}.$$

By comparison, the angular momentum of the Sun (assuming uniform rotation) is  $1.63 \times 10^{41} \text{ kg m}^2 \text{ s}^{-1}$ , about 18% of  $L_{\max}$ . We should expect that many stars will have angular momenta that are comparable to  $L_{\max}$ , and so vigorous (if not maximal) rotation ought to be common for stellar-mass black holes.

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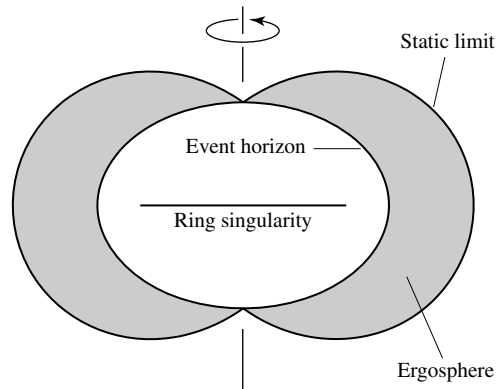
### Spacetime Frame Dragging

The structure of a maximally rotating black hole is shown in Fig. 17.22.<sup>23</sup> The rotation has distorted the central singularity from a point into a flat ring, and the event horizon has assumed the shape of an ellipsoid. The figure also shows additional features caused by the rotation. As a massive object spins, it induces a rotation in the surrounding spacetime, a phenomenon known as **frame dragging**. To gain some insight into this effect, recall the behavior of a pendulum swinging at the north pole of Earth. As Earth rotates, the plane of the pendulum’s swing remains fixed with respect to the distant stars. The stars define

<sup>21</sup>If magnetic monopoles exist, the “magnetic charge” would also be required for a complete specification. However, both magnetic and electric charge can be safely ignored because stars should be very nearly neutral.

<sup>22</sup>The “no hair” theorem actually applies only to the universe outside the event horizon. Inside, the spacetime geometry is complicated by the mass distribution of the collapsed star.

<sup>23</sup>The *Kerr metric* for a rotating black hole was derived from Einstein’s field equations by a New Zealand mathematician, Roy Kerr, in 1963.



**FIGURE 17.22** The structure of a maximally rotating black hole, with the ring singularity seen edge-on. The location of the event horizon at the equator is  $r = \frac{1}{2}R_S = GM/c^2$ .

a nonrotating frame of reference for the universe, and it is relative to this frame that the pendulum's swing remains planar. However, the rotating spacetime close to a massive spinning object produces a local deviation from the nonrotating frame that describes the universe at large. Near a rotating black hole, frame dragging is so severe that there is a nonspherical region outside the event horizon called the **ergosphere** where any particle *must* move in the same direction that the black hole rotates. Spacetime within the ergosphere is rotating so rapidly that a particle would have to travel faster than the speed of light to remain at the same angular coordinate (e.g., at the same value of  $\phi$  in the coordinate system used by a distant observer). The outer boundary of the ergosphere is called the **static limit**, so named because once beyond this boundary, a particle can remain at the same coordinate as the effect of frame dragging diminishes.

Even Earth's rotation produces very weak frame dragging. Detecting the effect of frame dragging was the mission of the Stanford Gravity Probe B experiment. The polar-orbit spacecraft was launched in April 2004 and ended data collection in October 2005 when the cryogenic liquid helium was used up. The experiment employed four superconducting gyroscopes made of precisely shaped spheres of fused quartz 3.8 cm in diameter. The gyroscopes were so nearly freely rotating that they formed an almost perfect spacetime reference frame. Although the predicted precession rate of the gyroscopes was only  $0.042'' \text{ yr}^{-1}$ , the effect of frame dragging is cumulative. It is anticipated that frame dragging will be measurable by comparing the changes that occurred in the gyroscopes' different initial orientations.<sup>24</sup>

At this point, you should be warned that the previous descriptions of a black hole's structure inside the event horizon, such as Fig. 17.22, are based on vacuum solutions to Einstein's field equations. These solutions were obtained by ignoring the effects of the mass of the collapsing star, so the vacuum solutions do not describe the interior of a real black hole. Furthermore, the present laws of physics, including general relativity, break down under

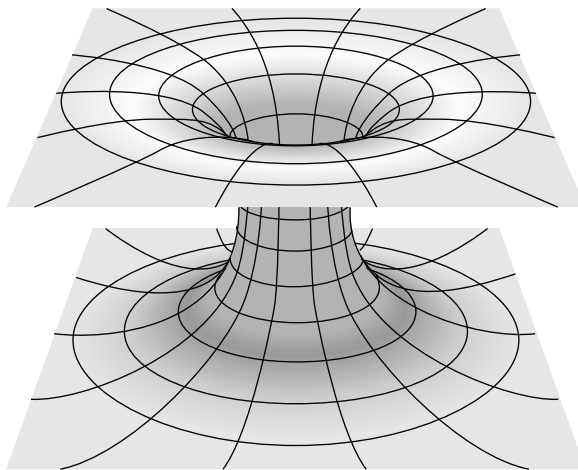
<sup>24</sup>At the time the text was written, the year-long, painstaking process of data analysis was under way. Results of the experiment are expected to be announced in late 2006 (for updates, visit the Stanford Gravity Probe B website at <http://einstein.stanford.edu/>).

the extreme conditions found very near the center. The details of the singularity cannot be fully described until a theory of quantum gravity is found. The presence of a singularity seems assured, however. In 1965 an English mathematician, Roger Penrose, proved that *every* complete gravitational collapse must form a singularity.

### Tunnels in Spacetime

The possibility of using a black hole as a tunnel connecting one location in spacetime with another (perhaps in a different universe) has inspired both physicists and science fiction writers. Most conjectures of spacetime tunnels are based on vacuum solutions to Einstein's field equations and as such don't apply to the interiors of real black holes. Still, they have become part of the popular culture, and we will consider them briefly here. Figure 17.23 depicts a spacetime tunnel called a **Schwarzschild throat** (also known as an *Einstein–Rosen bridge*), which uses the Schwarzschild geometry of a nonrotating black hole to connect two regions of spacetime. The width of the throat is a minimum at the event horizon, and the “mouths” may be interpreted as opening onto two different locations in spacetime. It is tempting to imagine this as a tunnel, and writers of speculative fiction have dreamed of *white holes* pouring out mass or serving as passageways for starships. However, it appears that any attempt to send a tiny amount of matter or energy (even a stray photon) through the throat would cause it to collapse. For a real nonrotating black hole, all worldlines end at the inescapable singularity, where spacetime is infinitely curved. There is simply no way to bypass the singularity.

The story is somewhat different for a rotating black hole. Although spacetime is still infinitely curved at the ring singularity, all worldlines need not converge there. In fact, it is difficult for an infalling object to hit the singularity in a rotating black hole. Theorists have calculated worldlines for vacuum solutions that miss the singularity and emerge in the spacetime of another universe. But just as for nonrotating black holes, any attempt to pass



**FIGURE 17.23** Depiction of a Schwarzschild throat connecting two different regions of spacetime. Any attempted passage of matter or energy through the throat would cause it to collapse.

the smallest amount of matter or energy along such a route would cause the passageway to collapse, thereby pinching it off. In summary, it seems extremely unlikely that black holes can provide a stable passageway for any matter or energy, even for idealized cases. For more realistic situations, any voyager attempting a trip through a black hole would end up being torn apart by the singularity.

Another possibility is that of a **wormhole**, a hypothetical tunnel between two points in spacetime separated by an arbitrarily great distance.<sup>25</sup> We will briefly consider nonrotating, spherically symmetric wormholes. They are described by *nonvacuum* solutions to Einstein's field equation. In other words, a wormhole must be threaded by some sort of *exotic material* whose tension prevents the collapse of the wormhole. There is no known mechanism that would allow a wormhole to arise naturally; it would have to be constructed by an incredibly advanced civilization. However, the theoretical possibilities alone are fascinating. These solutions to Einstein's field equations have no event horizon (permitting two-way trips through the wormhole) and involve survivable tidal forces. Journey times from one end through to the other can be less than one year (traveler's proper time), although the ends of the wormhole may be separated by interstellar or intergalactic distances.

The catch, of course, is the problematic existence of the exotic material needed to stabilize the wormhole. The unusual nature of the exotic material becomes apparent if we consider two light rays that converge on the wormhole and enter it, only to diverge when they exit the other end. This implies that the exotic material must be capable of gravitationally defocusing light, an "antigravity" effect involving the gravitational repulsion of the light by the material through which the rays pass. Exotic material meeting this requirement would have a negative energy density ( $\rho c^2 < 0$ ), at least as experienced by the light rays. Although a negative energy density arises in certain quantum situations, it may or may not be allowed physically on macroscopic scales. We will leave wormholes as a fascinating possibility and abandon the discussion at this point, recalling Einstein's remark that "all our thinking is of the nature of a free play with concepts."<sup>26</sup>

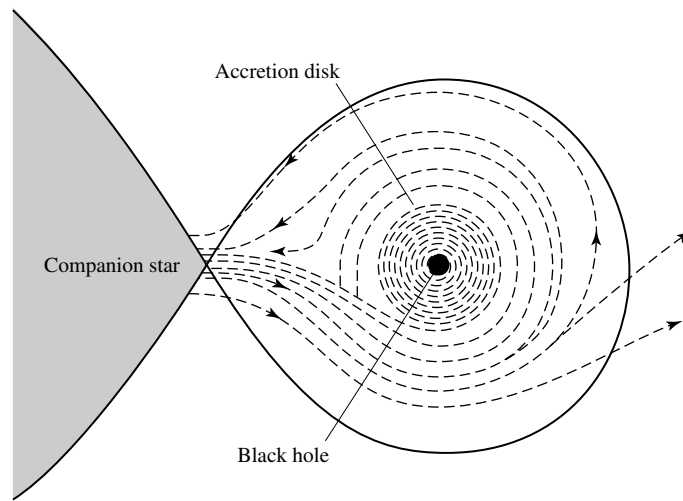
### Stellar-Mass Black Hole Candidates

You may feel as though much of this section was borrowed from the pages of a science fiction novel. Extraordinary claims require extraordinary proof, and proof of the mere existence of black holes has been difficult to obtain. The problem lies in detecting an object only a few tens of kilometers across that emits no radiation directly. The best hope of astronomers has been to find a black hole in a close binary system. If the black hole in such a system is able to pull gas from the envelope of the normal companion star, the angular momentum of their orbital motion would cause a disk of gas to form around the black hole (see Fig. 17.24).<sup>27</sup> As the gas spirals down toward the event horizon, it is compressed and heated to millions of kelvins and emits X-rays. Only the gravity of a neutron star or a black hole can produce X-rays in a close binary system, and in fact the compact object in most X-ray binaries is

<sup>25</sup>The term *wormhole* recalls the holes eaten in some ancient map books by worms, providing a symbolic shortcut between the distant locations portrayed on the maps.

<sup>26</sup>You are referred to Morris and Thorne (1988) and Thorne (1994) for more details and further speculations concerning wormholes.

<sup>27</sup>Accretion disks in a close binary system will be discussed in more detail in Section 18.2.



**FIGURE 17.24** Gas pulled from a companion star forms an X-ray emitting disk around a black hole.

believed to be a neutron star. However, if an X-ray binary can be found in which the mass of the compact object exceeds  $3 M_{\odot}$ , then a strong case can be made that the compact object is a black hole.

The first black hole to be tentatively identified in this way is Cygnus X-1, near the bright star  $\eta$  Cygni in the middle of the swan's neck. Another promising candidate is LMC X-3, an X-ray binary in the Large Magellanic Cloud. Yet another compelling case involves the X-ray binary A0620–00, also known as V616 Mon.<sup>28</sup> The orbital velocities along the line of sight of *both* components of A0620–00 have been determined using Doppler-shifted spectral lines. A simple application of Kepler's laws shows that the mass of the compact object must be *at least*  $3.82 \pm 0.24 M_{\odot}$ , well above the  $3 M_{\odot}$  upper limit for a neutron star.

However, perhaps the strongest candidate is V404 Cygni. Originally determined to be a recurrent nova (see Section 18.4), V404 Cyg underwent an X-ray outburst in 1989 that was detected by the Ginga satellite. Examination with ground-based optical telescopes revealed a companion star of type K0 IV with a radial velocity amplitude of  $211 \pm 4 \text{ km s}^{-1}$  and an orbital period of  $6.473 \pm 0.001 \text{ d}$ . The best estimate of the mass of the unseen companion to the K0 IV star is  $12 \pm 2 M_{\odot}$ . As more evidence accumulates, it seems that astronomers have finally found the extraordinary proof required for the existence of a black hole.

### Hawking Radiation

The black holes of classical general relativity last forever. A very general result derived by Stephen Hawking states that the surface area of a black hole's event horizon can never decrease. If a black hole coalesces with any other object, the result is an even larger black

<sup>28</sup>A0620–00 (V616 Mon) is on the border of the constellations Monoceros and Orion, about one-third of the way along a line from Betelgeuse to Sirius.

hole. In 1974, however, Hawking discovered a loophole in this law when he combined quantum mechanics with the theory of black holes and found that black holes can slowly *evaporate*. The key to this process is pair production, the formation of a particle–antiparticle pair just outside the event horizon of a black hole. Ordinarily the particles quickly recombine and disappear, but if one of the particles falls into the event horizon while its partner escapes, as shown in Fig. 17.25, this disappearing act may be thwarted. The black hole’s gravitational energy was used to produce the two particles, and so the escaping particle has carried away some of the black hole’s mass. The net effect as seen by an observer at a great distance is the emission of particles by the black hole, known as **Hawking radiation**, accompanied by a reduction in the black hole’s mass.

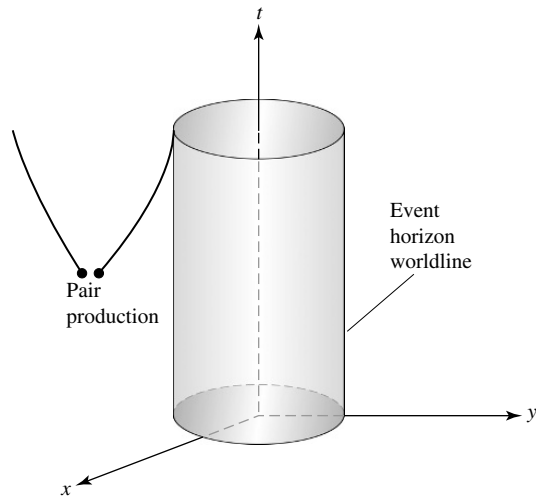
The rate at which energy is carried away by particles in this manner is inversely proportional to the square of the black hole’s mass, or  $1/M^2$ . For stellar-mass black holes, the emitted particles are photons and the rate of emission is minuscule. As the black hole’s mass declines, however, the rate of emission increases. The final stage of a black hole’s evaporation proceeds extremely rapidly, releasing a burst of all types of elementary particles. This tremendous explosion probably leaves behind only an empty region of flat spacetime.

The lifetime of a primordial black hole prior to its evaporation,  $t_{\text{evap}}$ , is quite long,

$$t_{\text{evap}} = 2560\pi^2 \left( \frac{2GM}{c^2} \right)^2 \left( \frac{M}{h} \right) \quad (17.30)$$

$$\approx 2 \times 10^{67} \left( \frac{M}{M_{\odot}} \right)^3 \text{ yr.}$$

Since the age of the universe is 13.7 billion years, this process is of no consequence for black holes formed by a collapsing star. However, a primordial black hole with a mass of



**FIGURE 17.25** Spacetime diagram showing particle–antiparticle pairs created near the event horizon of a black hole.

roughly  $1.7 \times 10^{11}$  kg would evaporate in about 13 billion years. Thus primordial black holes with this mass should be in the final, explosive stage of evaporation right now and could possibly be detected. The final burst of Hawking radiation is thought to release high-energy ( $\approx 100$  MeV) gamma rays at a rate of  $10^{13}$  W, together with electrons, positrons, and many other particles. The subsequent decay of these particles should produce additional gamma rays that would be observable by Earth-orbiting satellites. To date, measurements of the cosmic gamma-ray background at this energy have not detected anything that can be identified with the demise of a nearby primordial black hole. Although there is as yet no positive evidence that primordial black holes exist, this negative result is still important. It implies that on average there cannot be more than 200 primordial black holes with this mass in every cubic light-year of space.

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### Technical

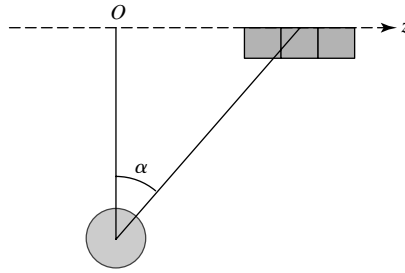
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## PROBLEMS

- 17.1** In the rubber sheet analogy of Section 17.1, a keen eye would notice that the tennis ball also depresses the sheet slightly, and so the bowling ball constantly tilts slightly toward the tennis ball as they orbit each other. Qualitatively compare this with the motion of two stars in a binary orbit.
- 17.2** Show that Eq. (17.6) for the gravitational redshift remains valid even if the light travels upward at an angle  $\theta$  measured from the vertical as long as  $h$  is taken to be the *vertical distance* traveled by the light pulse.
- 17.3** A photon near the surface of Earth travels a horizontal distance of 1 km. How far does the photon “fall” in this time?
- 17.4** Leadville, Colorado, is at an altitude of 3.1 km above sea level. If a person there lives for 75 years (as measured by an observer at a great distance from Earth), how much longer would gravitational time dilation have allowed that person to live if he or she had moved at birth from Leadville to a city at sea level?
- 17.5** (a) Estimate the radius of curvature of a horizontally traveling photon at the surface of a  $1.4 M_{\odot}$  neutron star, and compare the result with the 10-km radius of the star. Can general relativity be neglected when studying neutron stars?
- (b) If one hour passes at the surface of the neutron star, how much time passes at a great distance? Compare the times obtained from the exact and approximate expressions, Eqs. (17.13) and (17.14), respectively.



**FIGURE 17.26** Local inertial frames for measuring the deflection of light near the Sun.

**17.6** Imagine a series of rectangular local inertial reference frames suspended by cables in a line near the Sun's surface, as shown in Fig. 17.26. The frames are carefully lined up so that the tops and sides of neighboring frames are parallel, and the tops of the frames lie along the  $z$ -axis. A photon travels unhindered through the frames. As the photon enters each frame, the frame is released from rest and falls freely toward the center of the Sun.

- (a) Show that as it passes through the frame located at angle  $\alpha$  (shown in the figure), the angular deflection of the photon's path is

$$d\phi = \frac{g_0 \cos^3 \alpha}{c^2} dz,$$

where  $dz$  is the width of the reference frame and  $g_0$  is the Newtonian gravitational acceleration at the point of closest approach,  $O$ . The angular deflection is small, so assume that the photon is initially traveling in the  $z$ -direction as it enters the frame. (*Hint:* The width of the frame in the  $z$ -direction is  $dz$ , so the time for the photon to cross the frame can be taken to be  $dz/c$ .)

- (b) Integrate the result you found in part (a) from  $\alpha = -\pi/2$  to  $+\pi/2$  and so find the total angular deflection of the photon as it passes through the curved spacetime near the Sun.
- (c) Your answer (which is also the answer obtained by Einstein in 1911 before he arrived at his field equations) is only half the correct value of  $1.75''$ . Can you *qualitatively* account for the missing factor of two?

**17.7** Assume that you are at the origin of a laboratory reference system at time  $t = 0$  when you start your clock (event  $A$ ). Determine whether the following events are within the future light cone of event  $A$ , within the past light cone of event  $A$ , or elsewhere.

- (a) A flashbulb goes off 7 m away at time  $t = 0$ .
- (b) A flashbulb goes off 7 m away at time  $t = 2$  s.
- (c) A flashbulb goes off 70 km away at time  $t = 2$  s.
- (d) A flashbulb goes off 700,000 km away at time  $t = 2$  s.
- (e) A supernova explodes 180,000 ly away at time  $t = -5.7 \times 10^{12}$  s.
- (f) A supernova explodes 180,000 ly away at time  $t = 5.7 \times 10^{12}$  s.
- (g) A supernova explodes 180,000 ly away at time  $t = -5.6 \times 10^{12}$  s.
- (h) A supernova explodes 180,000 ly away at time  $t = 5.6 \times 10^{12}$  s.

For items (e) and (g), could an observer in another reference frame moving relative to yours measure that the supernova exploded *after* event  $A$ ? For items (f) and (h), could an observer in another frame measure that the supernova exploded *before* event  $A$ ?

- 17.8**  $\tau$  Ceti is the closest single star that is similar to the Sun. At time  $t = 0$ , Alice leaves Earth in her starship and travels at a speed of  $0.95c$  to  $\tau$  Ceti, 11.7 ly away as measured by astronomers on Earth. Her twin brother, Bob, remains at home, at  $x = 0$ .
- (a) According to Bob, what is the interval between Alice's leaving Earth and arriving at  $\tau$  Ceti?
  - (b) According to Alice, what is the interval between her leaving Earth and arriving at  $\tau$  Ceti?
  - (c) Upon arriving at  $\tau$  Ceti, Alice immediately turns around and returns to Earth at a speed of  $0.95c$ . (Assume that the actual turnaround takes negligible time.) What was the proper time for Alice during her round trip to  $\tau$  Ceti?
  - (d) When she and Bob meet on her return to Earth, how much younger will Alice be than her brother?

- 17.9** Consider a spherical blackbody of constant temperature and mass  $M$  whose surface lies at radial coordinate  $r = R$ . An observer located at the surface of the sphere and a distant observer both measure the blackbody radiation given off by the sphere.

- (a) If the observer at the surface of the sphere measures the luminosity of the blackbody to be  $L$ , use the gravitational time dilation formula, Eq. (17.13), to show that the observer at infinity measures

$$L_{\infty} = L \left( 1 - \frac{2GM}{Rc^2} \right). \quad (17.31)$$

- (b) Both observers use Wien's law, Eq. (3.15), to determine the blackbody's temperature. Show that

$$T_{\infty} = T \sqrt{1 - \frac{2GM}{Rc^2}}. \quad (17.32)$$

- (c) Both observers use the Stefan–Boltzmann law, Eq. (3.17), to determine the radius of the spherical blackbody. Show that

$$R_{\infty} = \frac{R}{\sqrt{1 - 2GM/Rc^2}}. \quad (17.33)$$

Thus, using the Stefan–Boltzmann law without including the effects of general relativity will lead to an *overestimate* of the size of a compact blackbody.

- 17.10** In 1792 the French mathematician Simon-Pierre de Laplace (1749–1827) wrote that a hypothetical star, “of the same density as Earth, and whose diameter would be two hundred and fifty times larger than the Sun, would not, in consequence of its attraction, allow any of its rays to arrive at us.” Use Newtonian mechanics to calculate the escape velocity of Laplace's star.
- 17.11** Qualitatively describe the effects on the orbits of the planets if the Sun were suddenly to become a black hole.
- 17.12** Consider four black holes with masses of  $10^{12}$  kg,  $10 M_{\odot}$ ,  $10^5 M_{\odot}$ , and  $10^9 M_{\odot}$ .
- (a) Calculate the Schwarzschild radius for each.
  - (b) Calculate the average density, defined by  $\rho = M/(\frac{4}{3}\pi R_s^3)$ , for each.
- 17.13** (a) Show that the proper distance from the event horizon to a radial coordinate  $r$  is given by

$$\Delta\mathcal{L} = r\sqrt{1 - \frac{R_s}{r}} + \frac{R_s}{2} \ln \left( \frac{1 + \sqrt{1 - R_s/r}}{1 - \sqrt{1 - R_s/r}} \right).$$

This illustrates the danger of interpreting  $r$  as a distance instead of a coordinate. *Hint:* Integrate Eq. (17.23).

- (b) Make a graph of  $\Delta\mathcal{L}$  as a function of  $r$  for values of  $r$  between  $r = R_S$  and  $r = 10R_S$ .
- (c) Show that, for large values of  $r$ ,

$$\Delta\mathcal{L} \simeq r.$$

Thus, far from the black hole, the radial coordinate  $r$  can be treated as a distance.

- 17.14** Verify that the area of the event horizon of a black hole is  $4\pi R_S^2$ . (*Hint:* Remember that the radial coordinate  $r$  is *not* the distance to the center. Use the Schwarzschild metric as your starting point.)
- 17.15** Equation (17.26) describes the coordinate speed of a massive particle orbiting a nonrotating black hole. However, it can be shown that the orbit is not stable unless  $r \geq 3R_S$ ; any disturbance will cause a particle in a smaller orbit to spiral down to the event horizon.
- (a) Find the coordinate speed of a particle in the smallest stable orbit around a  $10 M_\odot$  black hole.
  - (b) Find the orbital period (in coordinate time  $t$ ) for this smallest stable orbit around a  $10 M_\odot$  black hole.
- 17.16** (a) Find an expression for the coordinate speed of light in the  $\phi$ -direction.  
 (b) Consider Eq. (17.26) in the limit that the particle's mass goes to zero and its speed approaches that of light. Use your result for part (a) to show that  $r = 1.5R_S$  for the circular orbit of a photon around a black hole.  
 (c) Find the orbital period (in coordinate time  $t$ ) for this orbit around a  $10 M_\odot$  black hole.  
 (d) If a flashlight were beamed in the  $\phi$ -direction at  $r = 1.5R_S$ , what would happen? (The surface at  $r = 1.5R_S$  is called the **photon sphere**.)
- 17.17** To obtain a crude estimate for the maximum angular momentum of a rotating black hole, imagine (obviously incorrectly!) that the black hole's mass is distributed uniformly throughout a solid sphere of radius  $R_S$  (the Schwarzschild radius). From basic physics, the moment of inertia of a uniform, rotating sphere is  $I = \frac{2}{5}MR^2$ , and the angular momentum of the sphere is  $L = I\omega$ , where  $\omega$  is the sphere's angular velocity. From this classical approach, estimate the maximum angular momentum of the solid sphere, and compare your answer with Eq. (17.29). (Be sure to specify any additional assumptions you have made.) What is the percentage error in your estimate compared to the exact result?
- 17.18** Use Eq. (17.29) to compare the maximum angular momentum of a  $1.4 M_\odot$  black hole with the angular momentum of the fastest known pulsar, which rotates with a period of 0.00139 s. Assume that the pulsar is a  $1.4 M_\odot$  uniform sphere of radius 10 km.
- 17.19** An electron is a point-like particle of zero radius, so it is natural to wonder whether an electron could be a black hole. However, a black hole of mass  $M$  cannot have an arbitrary amount of angular momentum  $L$  and charge  $Q$ . These values must satisfy an inequality,

$$\left(\frac{GM}{c}\right)^2 \geq G \left(\frac{Q}{c}\right)^2 + \left(\frac{L}{M}\right)^2.$$

If this inequality were violated, the singularity would be found *outside* the event horizon, in violation of the Law of Cosmic Censorship. Use  $\hbar/2$  for the electron's angular momentum to determine whether or not an electron is a black hole.

- 17.20 (a)** The angular rotation rate,  $\Omega$ , at which spacetime is dragged around a rotating mass must be proportional to its angular momentum  $L$ . The expression for  $\Omega$  must also contain the constants  $G$  and  $c$ , together with the radial coordinate  $r$ . Show on purely dimensional grounds that

$$\Omega = \text{constant} \times \frac{GL}{r^3 c^2},$$

where the constant (which you need not determine) is of order unity.

- (b) Evaluate this for Earth, assuming that it is a uniformly rotating sphere. Set the leading constant equal to one, and express your answer in arcseconds per year. How much time would it take for a pendulum at the north pole to rotate *once* relative to the distant stars because of frame dragging?
- (c) Repeat part (b) for the fastest known pulsar, expressing  $\Omega$  in revolutions per second.
- 17.21 (a)** Use dimensional arguments to combine the fundamental constants  $\hbar$ ,  $c$ , and  $G$  into an expression that has units of mass. Evaluate your result, which is an estimate of the least massive primordial black hole formed in the first instant after the Big Bang. What is the mass in kilograms?
- (b) What is the Schwarzschild radius for such a black hole?
- (c) How long would it take light to travel this distance?
- (d) What is the lifetime of this black hole before its evaporation?
- 17.22** By combining gravitation ( $G$ ), thermodynamics ( $k$ ), and quantum mechanics ( $\hbar$ ), Stephen Hawking calculated the temperature,  $T$ , of a nonrotating black hole to be

$$kT = \frac{\hbar c^3}{8\pi G M} = \frac{\hbar c}{4\pi R_S}, \quad (17.34)$$

where  $R_S$  is the Schwarzschild radius.

- (a) Verify that the expression has the right units.
- (b) It was mentioned in the text that if a primordial black hole formed at the beginning of the universe 13.7 Gyr ago with a mass of  $1.7 \times 10^{11}$  kg, it would be reaching the end of its life now. Compute the temperature of a primordial black hole having a mass of  $1.7 \times 10^{11}$  kg.
- (c) Approximately what portion of the electromagnetic spectrum would this blackbody temperature correspond to?
- (d) What would the radius of a sphere having the density of water be if it had a mass of  $1.7 \times 10^{11}$  kg?
- (e) Compute the temperature of a  $10 M_\odot$  black hole.
- 17.23** In this problem we will derive the expression for the lifetime of a nonrotating, evaporating black hole (Eq. 17.30).
- (a) Consider a black hole to be a perfectly radiating blackbody of temperature  $T$ , given by Eq. (17.34) in Problem 17.22. Assuming that the surface area of the black hole is given by  $4\pi R_S^2$ , where  $R_S$  is the Schwarzschild radius, show that the luminosity of the black hole due to Hawking radiation is

$$L = \frac{\hbar c^6}{15360\pi G^2 M^2} = \frac{\hbar c^2}{3840\pi R_S^2}.$$

- (b) The luminosity of the black hole must originate from a loss in the black hole's internal energy. Assuming that the energy of the black hole is given by  $E = Mc^2$  and that  $L = dE/dt$ , show that the time required for the black hole to lose all of its mass to Hawking radiation is given by Eq. (17.30).

**17.24** In the X-ray binary system A0620–00, the radial orbital velocities for the normal star and the compact object are  $v_{s,r} = 457 \text{ km s}^{-1}$  and  $v_{c,r} = 43 \text{ km s}^{-1}$ , respectively. The orbital period is 0.3226 day.

- (a) Calculate the mass function (the right-hand side of Eq. 7.7),

$$\frac{m_c^3}{(m_s + m_c)^2} \sin^3 i,$$

where  $m_s$  is the mass of the normal star,  $m_c$  is the mass of its compact companion, and  $i$  is the angle of inclination of the orbit. What does this result say about the mass of the compact object? (Note that the value of  $v_{c,r}$  was not needed to obtain this result.)

- (b) Now use the value of the orbital radial velocity of the compact object to determine its mass, assuming  $i = 90^\circ$ . What does this result say about the mass of the compact object?
- (c) The X-rays are not eclipsed in this system, so the angle of inclination must be less than approximately  $85^\circ$ . Suppose that the angle of inclination were  $45^\circ$ . What would the mass of the compact object be then?

**17.25** From the data in the text, make an estimate of the lower mass limit of the compact companion in V404 Cyg by setting the orbital inclination  $i = 90^\circ$ . Refer to Appendix G in order to make a rough determination of the mass of the K0 IV companion. Why does your calculation result in a lower limit? What further information would be required to make a more precise determination?

## Close Binary Star Systems

- 18.1 *Gravity in a Close Binary Star System*
- 18.2 *Accretion Disks*
- 18.3 *A Survey of Interacting Binary Systems*
- 18.4 *White Dwarfs in Semidetached Binaries*
- 18.5 *Type Ia Supernovae*
- 18.6 *Neutron Stars and Black Holes in Binaries*

**18.1 ■ GRAVITY IN A CLOSE BINARY STAR SYSTEM**

As explained in Chapter 7, at least half of all “stars” in the sky are actually multiple systems, consisting of two (or more) stars in orbit about their common center of mass. In most of these systems the stars are sufficiently far apart that they have a negligible impact on one another. They evolve essentially independently, living out their lives in isolation except for the gentle grip of gravity that binds them together.

If the stars are very close, with a separation roughly equal to the diameter of the larger star, then one or both stars may have their outer layers gravitationally deformed into a teardrop shape. As a star rotates through the tidal bulge raised by its partner’s gravitational pull, it is forced to pulsate. These oscillations are damped by the mechanisms discussed in Section 14.2. Orbital and rotational energy is dissipated in this way until the system reaches the state of minimum energy for its (constant) angular momentum, resulting in synchronous rotation and circular orbits. Thereafter the same side of each star always faces the other as the system rotates rigidly in space, and no further energy can be lost by tidally driven oscillations.<sup>1</sup> The distorted star may even lose some of its photospheric gases to its companion. The spilling of gas from one star onto another can lead to some spectacular celestial fireworks, the subject of this chapter.

**Lagrangian Points and Equipotential Surfaces**

To understand how gravity operates in a close binary star system, consider two stars in a circular orbit in the  $x$ - $y$  plane with angular velocity  $\omega = v_1/r_1 = v_2/r_2$ . Here,  $v_1$  and  $r_1$  are the orbital speed of Star 1 and its distance from the center of mass of the system, and similarly for Star 2. It is useful to choose a corotating coordinate system that follows the

<sup>1</sup>If one of the stars is a compact object such as a white dwarf or a neutron star, its spin may not be synchronized.

rotation of the two stars about their center of mass. If the center of mass is at the origin, then the stars will be at rest in this rotating reference frame, with their mutual gravitational attraction balanced by the outward “push” of a centrifugal force.<sup>2</sup> The centrifugal force vector on a mass  $m$  in this frame a distance  $r$  from the origin is then

$$\mathbf{F}_c = m\omega^2 r \hat{\mathbf{r}}, \quad (18.1)$$

in the outward radial direction.

It is usually easier to work with the gravitational potential energy, given by Eq. (2.14),

$$U_g = -G \frac{Mm}{r},$$

instead of with the gravitational force.<sup>3</sup> To do this in a rotating coordinate system, we must include a fictitious “centrifugal potential energy” in the potential energy term through the use of Eq. (2.13):

$$U_f - U_i = \Delta U_c = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F}_c \cdot d\mathbf{r}.$$

Here,  $\mathbf{F}_c$  is the centrifugal force vector,  $\mathbf{r}_i$  and  $\mathbf{r}_f$  are the initial and final position vectors, respectively, and  $d\mathbf{r}$  is the infinitesimal change in the position vector (recall Fig. 2.9). The change in centrifugal potential energy is thus

$$\Delta U_c = - \int_{r_i}^{r_f} m\omega^2 r dr = -\frac{1}{2}m\omega^2 (r_f^2 - r_i^2).$$

Realizing that only *changes* in potential energy are physically meaningful, we can arbitrarily choose  $U_c = 0$  at  $r = 0$  to give the final result for the centrifugal potential energy,

$$U_c = -\frac{1}{2}m\omega^2 r^2. \quad (18.2)$$

Figure 18.1 shows a corotating coordinate system in which two stars with masses  $M_1$  and  $M_2$  are separated by a distance  $a$ . The stars are located on the  $x$ -axis at distances  $r_1$  and  $r_2$ , respectively, from the center of mass, which is placed at the origin. Thus

$$r_1 + r_2 = a \quad \text{and} \quad M_1 r_1 = M_2 r_2. \quad (18.3)$$

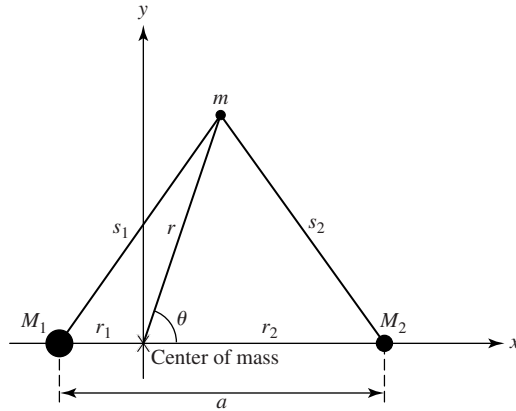
Including the centrifugal term, the effective potential energy for a small test mass  $m$  located in the plane of the orbit (the  $x$ - $y$  plane) is

$$U = -G \left( \frac{M_1 m}{s_1} + \frac{M_2 m}{s_2} \right) - \frac{1}{2}m\omega^2 r^2.$$

<sup>2</sup>The centrifugal force is an *inertial force* (as opposed to a physical force) that must be included when describing motion in a rotating coordinate system. There is another inertial force, called the *Coriolis force*, that will be neglected in what follows.

<sup>3</sup>Most stars can be treated as point masses in what follows because the mass is concentrated at their centers, allowing their teardrop shapes to be neglected.





**FIGURE 18.1** Corotating coordinates for a binary star system.

For convenience, the effective potential energy can be divided by  $m$  to obtain the **effective gravitational potential**,  $\Phi$ :

$$\Phi = -G \left( \frac{M_1}{s_1} + \frac{M_2}{s_2} \right) - \frac{1}{2} \omega^2 r^2. \quad (18.4)$$

This is just the effective potential energy *per unit mass*. From the law of cosines, the distances  $s_1$  and  $s_2$  are given by

$$s_1^2 = r_1^2 + r^2 + 2r_1 r \cos \theta \quad (18.5)$$

$$s_2^2 = r_2^2 + r^2 - 2r_2 r \cos \theta. \quad (18.6)$$

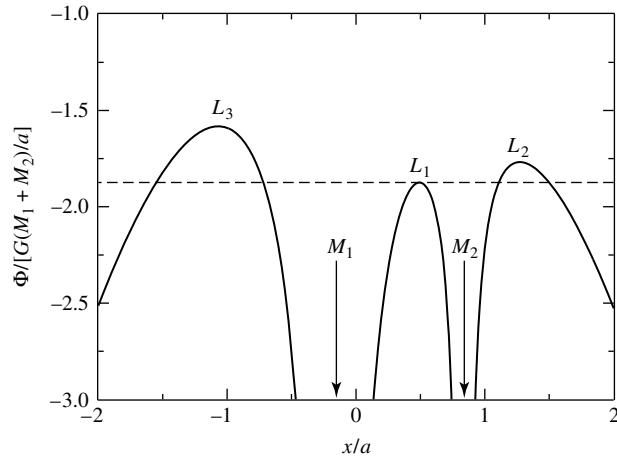
The angular frequency of the orbit,  $\omega$ , comes from Kepler's third law for the orbital period,  $P$ , Eq. (2.37),

$$\omega^2 = \left( \frac{2\pi}{P} \right)^2 = \frac{G(M_1 + M_2)}{a^3}. \quad (18.7)$$

Equations (18.3) and (18.4–18.7) can be used to evaluate the effective gravitational potential  $\Phi$  at every point in the orbital plane of a binary star system. For example, Fig. 18.2 shows the value of  $\Phi$  along the  $x$ -axis. The significance of this graph becomes clear when the  $x$ -component of the force on a small test mass  $m$ , initially at rest on the  $x$ -axis, is written as

$$F_x = -\frac{dU}{dx} = -m \frac{d\Phi}{dx} \quad (18.8)$$

(recall Eq. 2.15). The three “hilltops” labeled  $L_1$ ,  $L_2$ , and  $L_3$  are **Lagrangian points**, where there is no force on the test mass ( $d\Phi/dx = 0$ ). At these three equilibrium points, the



**FIGURE 18.2** The effective gravitational potential  $\Phi$  for two stars of mass  $M_1 = 0.85 M_\odot$ ,  $M_2 = 0.17 M_\odot$  on the  $x$ -axis. The stars are separated by a distance  $a = 5 \times 10^8 \text{ m} = 0.718 R_\odot$ , with their center of mass located at the origin. The  $x$ -axis is in units of  $a$ , and  $\Phi$  is expressed in units of  $G(M_1 + M_2)/a = 2.71 \times 10^{11} \text{ J kg}^{-1}$ . (In fact, the figure is the same for any  $M_2/M_1 = 0.2$ .) The dashed line is the value of  $\Phi$  at the inner Lagrangian point. If the total energy per unit mass of a particle exceeds this value of  $\Phi$ , it can flow through the inner Lagrangian point between the two stars.

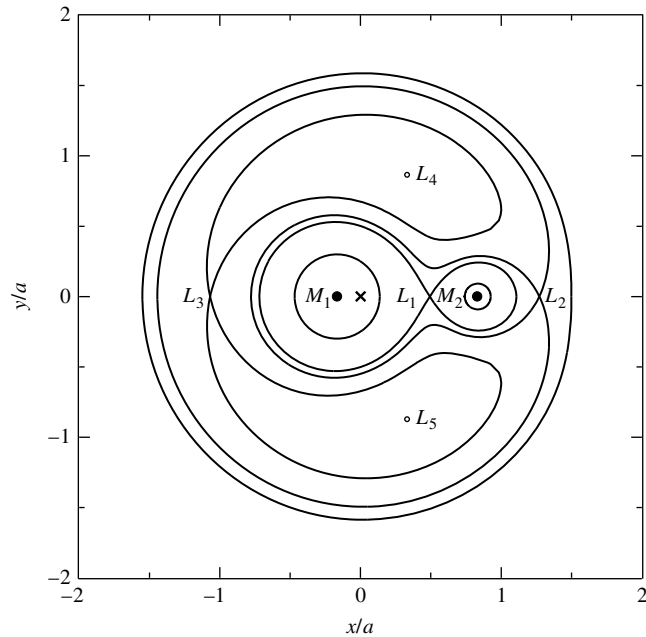
gravitational forces on  $m$  due to  $M_1$  and  $M_2$  are balanced by the centrifugal force.<sup>4</sup> These equilibrium points are *unstable* because they are *local maxima* of  $\Phi$ ; if the test mass is displaced slightly, the minus sign in Eq. (18.8) indicates that it will accelerate “downhill,” away from its equilibrium position. The *inner Lagrangian point*,  $L_1$ , plays a central role in close binary systems. Approximate expressions for the distances from  $L_1$  to  $M_1$  and  $M_2$ , denoted respectively by  $\ell_1$  and  $\ell_2$ , are

$$\ell_1 = a \left[ 0.500 - 0.227 \log_{10} \left( \frac{M_2}{M_1} \right) \right] \quad (18.9)$$

$$\ell_2 = a \left[ 0.500 + 0.227 \log_{10} \left( \frac{M_2}{M_1} \right) \right]. \quad (18.10)$$

Points in space that share the same value of  $\Phi$  form an **equipotential surface**. Figure 18.3 shows *equipotential contours* that outline the intersection of several equipotential surfaces with the plane of the orbit. Very close to either of the masses  $M_1$  or  $M_2$ , the equipotential surfaces are nearly spherical and centered on each mass. Farther away, the combined gravitational influence of  $M_1$  and  $M_2$  distorts the equipotential surfaces into teardrop shapes until

<sup>4</sup>From an inertial (nonrotating) frame of reference, this motion would be described by saying that gravitational forces of  $M_1$  and  $M_2$  produce the inward centripetal acceleration of the test mass as it orbits the center of mass of the system.



**FIGURE 18.3** Equipotentials for  $M_1 = 0.85 M_\odot$ ,  $M_2 = 0.17 M_\odot$ , and  $a = 5 \times 10^8 \text{ m} = 0.718 R_\odot$ . The axes are in units of  $a$ , with the system's center of mass (the “x”) at the origin. Starting at the top of the figure and moving down toward the center of mass, the values of  $\Phi$  in units of  $G(M_1 + M_2)/a = 2.71 \times 10^{11} \text{ J kg}^{-1}$  for the equipotential curves are  $\Phi = -1.875, -1.768, -1.583, -1.583, -1.768$  (the “dumbbell”),  $-1.875$  (the Roche lobe), and  $-3$  (the spheres).  $L_4$  and  $L_5$  are local maxima, with  $\Phi = -1.431$ .

they finally touch at the inner Lagrangian point. At even greater distances, the equipotential surfaces assume a “dumbbell” shape surrounding both masses.<sup>5</sup>

These equipotential surfaces are *level surfaces* for binary stars. In a binary system, as one of the stars evolves, it will expand to fill successively larger equipotential surfaces (somewhat like inflating a balloon). To see this, consider that the effective gravity at each point is always *perpendicular* to the equipotential surface there.<sup>6</sup> Hydrostatic equilibrium guarantees that the pressure is constant along a surface of constant  $\Phi$ ; there is no component of gravity parallel to an equipotential surface, and so a pressure difference in that direction

<sup>5</sup>We will not be concerned with the other equipotential contours on Fig. 18.3 that pass through the Lagrangian points  $L_3$ ,  $L_4$ , and  $L_5$ . However, the *Trojan asteroids* that accumulate at Lagrangian points  $L_4$  and  $L_5$ . (If  $M_1 > 24.96 M_2$ , as for the Sun and Jupiter, then the Coriolis force is strong enough to cause  $L_4$  and  $L_5$  to be *stable* equilibrium points.) Each of the Lagrangian points  $L_4$  and  $L_5$  forms an equilateral triangle with masses  $M_1$  and  $M_2$  in Fig. 18.3, so the Trojan asteroids are found at about  $60^\circ$  ahead of and behind Jupiter in its orbit, with a spread due to the finite width of the potential well.

<sup>6</sup>The mathematical statement of this is  $\mathbf{F} = -m\nabla\Phi$ . This is analogous to an electric field vector being oriented perpendicular to an electrical equipotential surface, pointing from higher to lower voltage.

cannot be balanced and maintained. And because the pressure is due to the weight of the overlying layers of the star, the density must also be the same along each equipotential surface in order to produce a constant pressure there.

### Classes of Binary Star Systems

The appearance of a binary star system depends on which equipotential surfaces are filled by the stars. Binary stars with radii much less than their separation are nearly spherical (as shown by the small circles in Fig. 18.3). This situation describes a **detached binary** in which the stars evolve nearly independently. Detached binary systems have already been described in Chapter 7 as a primary source of astronomical information about the basic properties of stars.

If one star expands enough to fill the “figure-eight” contour in Fig. 18.3, then its atmospheric gases can escape through the inner Lagrangian point  $L_1$  to be drawn toward its companion. The teardrop-shaped regions of space bounded by this particular equipotential surface are called **Roche lobes**.<sup>7</sup> The transfer of mass from one star to the other can begin when one of the stars has expanded beyond its Roche lobe. Such a system is called a **semidetached binary**. The star that fills its Roche lobe and loses mass is usually called the **secondary star**, with mass  $M_2$ , and its companion the **primary star** has mass  $M_1$ . The primary star may be either more or less massive than the secondary star.

It may happen that *both stars* fill, or even expand beyond, their Roche lobes. In this case, the two stars share a common atmosphere bounded by a dumbbell-shaped equipotential surface, such as the one passing through the Lagrangian point  $L_2$ . Such a system is called a **contact binary**. Figure 18.4 illustrates the three classes of binary stars.

### Mass Transfer Rate

A crude estimate of the rate at which mass is transferred in a semidetached binary may be obtained for the case of two stars of equal mass. Let the radius of the star that has expanded beyond its Roche lobe be  $R$ . The equipotential surface at the radius of this star will be modeled by two spheres of radius  $R$  that overlap slightly by a distance  $d$ , as shown in Fig. 18.5. We will assume that stellar gas will escape from the filled lobe through the circular opening of radius  $x$ . If the density of the stellar material at the opening is  $\rho$  and its speed toward the opening of area  $A = \pi x^2$  is  $v$ , then it is left as an exercise (Problem 18.3) to show that the rate at which mass leaves the filled lobe, the **mass transfer rate**, is

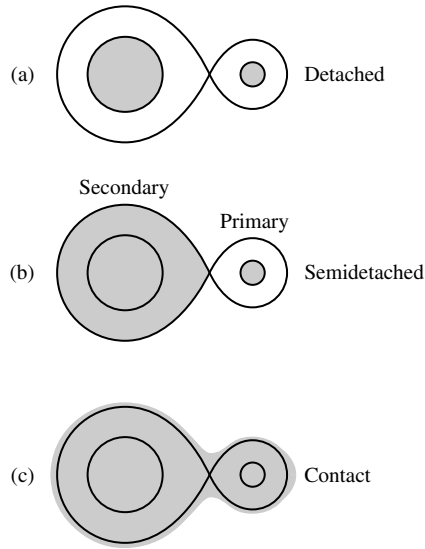
$$\dot{M} = \rho v A. \quad (18.11)$$

A bit of geometry shows that

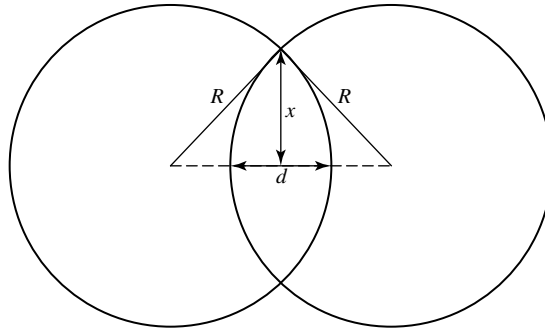
$$x = \sqrt{Rd} \quad (18.12)$$

when  $d \ll R$ . Using Eq. (8.3) for the thermal velocity of the gas particles results in the

<sup>7</sup>The term *Roche lobe* was chosen in honor of the nineteenth-century French mathematician Edouard Roche (1820–1883).



**FIGURE 18.4** The classification of binary star systems. (a) A detached system. (b) A semidetached system in which the secondary star has expanded to fill its Roche lobe. (c) A contact binary.



**FIGURE 18.5** Intersecting spheres used to estimate the mass transfer rate,  $\dot{M}$ .

estimate

$$\dot{M} \approx \rho v_{\text{rms}} \pi x^2 \quad (18.13)$$

or

$$\dot{M} \approx \pi R d \rho \sqrt{\frac{3kT}{m_H}}, \quad (18.14)$$

assuming a gas of hydrogen atoms. As the overfill distance  $d$  becomes larger, the values of the density and temperature increase at the opening. In Problem 18.32, you will be asked to

show that the mass transfer rate increases rapidly with  $d$ ; a more detailed calculation results in  $\dot{M} \propto d^3$ .

---

**Example 18.1.1.** Suppose a star like the Sun is in a semidetached binary system with a companion of equal mass, and it slightly overfills its Roche lobe to a point just below its photosphere. Using zone  $i = 9$  for the StatStar model in Table 9.5, we have  $d = r_0 - r_9 = 1.52 \times 10^6$  m,  $T = 6348$  K, and  $\rho = 2.87 \times 10^{-7}$  kg m $^{-3}$ . Using  $R = 7.10 \times 10^8$  m at the outermost point of the model, the rate at which this Sun-like star would lose its atmospheric gases would be roughly

$$\dot{M} \approx \pi R d \rho \sqrt{\frac{3kT}{m_H}} = 1.2 \times 10^{13} \text{ kg s}^{-1} = 1.9 \times 10^{-10} M_\odot \text{ yr}^{-1}.$$

This is typical of the mass transfer rates for semidetached binary systems. The values of  $\dot{M}$  inferred from observations of various systems range from  $10^{-11}$  to  $10^{-7} M_\odot \text{ yr}^{-1}$ . For comparison, recall from Example 11.2.1 that the solar wind transports mass away from the Sun at a much smaller rate, approximately  $3 \times 10^{-14} M_\odot \text{ yr}^{-1}$ .

---

Before moving on to consider the consequences of the transfer of mass in semidetached binaries, it is worthwhile to consider the enormous energy that can be released when matter falls onto a star, especially onto a compact object such as a white dwarf or a neutron star.

---

**Example 18.1.2.** Consider a mass  $m = 1$  kg that starts at rest infinitely far from a star of mass  $M$  and radius  $R$ . The initial total mechanical energy of the mass  $m$  is

$$E = K + U = 0.$$

Using conservation of energy, we find that the kinetic energy of the mass when it arrives at the star's surface is

$$K = -U = G \frac{Mm}{R}.$$

This kinetic energy will be converted into heat and light upon impact with the star. If the star is a white dwarf with  $M = 0.85 M_\odot$  and  $R = 6.6 \times 10^6$  m =  $0.0095 R_\odot$ , then the energy released by one kilogram of infalling matter is

$$G \frac{Mm}{R} = 1.71 \times 10^{13} \text{ J}.$$

This is 0.019% of the rest energy ( $mc^2$ ) of one kilogram of material. For comparison, the amount of energy released by the thermonuclear fusion of one kilogram of hydrogen is

$$0.007mc^2 = 6.29 \times 10^{14} \text{ J}$$

(recall Example 10.3.2).

If the star is a neutron star with mass  $M = 1.4 M_{\odot}$  and radius  $R = 10$  km, then the energy released is much greater:

$$G \frac{Mm}{R} = 1.86 \times 10^{16} \text{ J.}$$

This is 21% of the rest energy of one kilogram, nearly 30 times greater than the energy that hydrogen fusion could provide! The calculations show that infalling matter is capable of generating immense amounts of energy.

Observations of celestial X-ray sources have revealed objects with a steady X-ray luminosity of approximately  $10^{30}$  W. If this radiation were produced by gases pulled from a companion star that then fell onto a neutron star's surface, the amount of mass per second transferred between the two stars that would be needed to account for the observed luminosity is

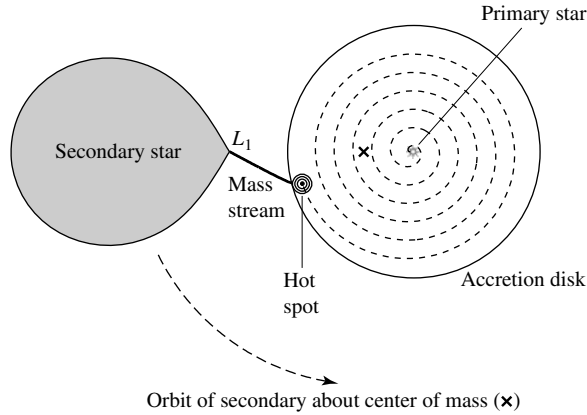
$$\dot{M} = \frac{10^{30} \text{ W}}{1.86 \times 10^{16} \text{ J kg}^{-1}} = 5.38 \times 10^{13} \text{ kg s}^{-1},$$

which is only about  $10^{-9} M_{\odot} \text{ yr}^{-1}$ . This is similar to the mass transfer rate found in the previous example, a fortuitous agreement because  $\dot{M}$  for semidetached systems can vary by several orders of magnitude.

## 18.2 ■ ACCRETION DISKS

The orbital motion of a semidetached binary can prevent the mass that escapes from the swollen secondary star from falling directly onto the primary star. The primary's movement is often enough to keep it out of the path of the gases that spill through the inner Lagrangian point. If the radius of the primary star is less than about 5% of the binary separation  $a$ , the mass stream will miss striking the primary's surface. Instead, the mass stream goes into orbit around the primary to form a thin **accretion disk** of hot gas in the orbital plane, as shown in Figs. 17.24 and 18.6.<sup>8</sup> **Viscosity**, an internal friction that converts the directed kinetic energy of bulk mass motion into random thermal motion, causes the orbiting gases to lose energy and slowly spiral inward toward the primary. The physical mechanism responsible for the viscosity in accretion disks is as yet poorly understood. The familiar molecular viscosity due to interparticle forces is far too weak to be effective. Other possibilities involve random motions of the gas, such as turbulence in the disk material caused by thermal convection or by a magnetohydrodynamic instability in the magnetic fields that interact with the differentially rotating disk (cf., Section 11.2). Whatever the mechanism, the gas is heated throughout its descent to increasingly higher temperatures as the lost orbital energy is converted into thermal energy. Finally, the plunging gas ends its journey at the star's surface.

<sup>8</sup>Astronomers refer to the process of accumulating mass from an outside source as *accretion*.



**FIGURE 18.6** A semidetached binary showing the accretion disk around the primary star and the hot spot where the mass streaming through the inner Lagrangian point impacts the disk. This system's parameters correspond to those of Z Chamaeleontis, described in Example 18.4.1.

### The Temperature Profile and Luminosity

Just as a star may be treated as a blackbody in a rough first approximation, the assumption of an optically thick accretion disk radiating as a blackbody provides a simple, useful model. At each radial distance, an optically thick disk emits blackbody radiation with a continuous spectrum corresponding to the local disk temperature at that distance.

To estimate the temperature of a model accretion disk at a distance  $r$  from the center of the primary star of mass  $M_1$  and radius  $R_1$ , let's assume that the inward radial velocity of the disk gases is small compared with their orbital velocity. Then, to a good approximation, the gases follow circular Keplerian orbits, and the details of the viscous forces acting within the disk may be neglected. Furthermore, since the mass of the disk is very small compared with that of the primary, the orbiting material feels only the gravity of the central primary star. The *total energy* (kinetic plus potential) of a mass  $m$  of orbiting gas is given by Eq. (2.35),

$$E = -G \frac{M_1 m}{2r}.$$

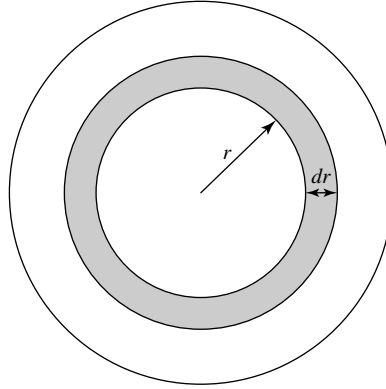
As the gas spirals inward, its total energy  $E$  becomes more negative. The lost energy maintains the disk's temperature and is ultimately emitted in the form of blackbody radiation.

Now consider an annular ring of radius  $r$  and width  $dr$  within the disk, as shown in Fig. 18.7. If the rate at which mass is transferred from the secondary to the primary star is a constant  $\dot{M}$ , then in time  $t$  the amount of mass that passes through the outer boundary of the circular ring shown in Fig. 18.7 is  $\dot{M}t$ . Assuming a *steady-state disk* that does not change with time, no mass is allowed to build up within the ring. Therefore during this time an amount of mass  $\dot{M}t$  must also leave through the ring's inner boundary.

Conservation of energy requires that the energy  $dE$  radiated by the ring in time  $t$  be equal to the difference in the energy that passes through the ring's outer and inner boundaries:

$$dE = \frac{dE}{dr} dr = \frac{d}{dr} \left( -G \frac{M_1 m}{2r} \right) dr = G \frac{M_1 \dot{M} t}{2r^2} dr,$$





**FIGURE 18.7** One of the (imaginary) annular rings constituting the accretion disk.

where  $m = \dot{M}t$  has been used for the orbiting mass entering and leaving the ring. If the luminosity of the ring is  $dL_{\text{ring}}$ , then the energy radiated by the ring in time  $t$  is related to  $dL_{\text{ring}}$  by

$$dL_{\text{ring}}t = dE = G \frac{M_1 \dot{M}t}{2r^2} dr.$$

Canceling the  $t$ 's and using the Stefan–Boltzmann law in the form of Eq. (3.16) with  $A = 2(2\pi r dr)$  for the surface area of the ring (both sides) gives

$$dL_{\text{ring}} = 4\pi r \sigma T^4 dr \quad (18.15)$$

$$= G \frac{M_1 \dot{M}}{2r^2} dr \quad (18.16)$$

for the luminosity of the ring. Solving for  $T$ , the disk temperature at radius  $r$ , results in

$$T = \left( \frac{GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} \left( \frac{R}{r} \right)^{3/4}. \quad (18.17)$$

The “1” subscript has been dropped, with the understanding that  $M$  and  $R$  are the mass and radius of the primary star, and that  $\dot{M}$  is the mass transfer rate for the semidetached binary system.

A more thorough analysis would take into account the thin turbulent boundary layer that must be produced when the rapidly orbiting disk gases encounter the surface of the primary star. This results in a better estimate of the disk temperature:

$$T = \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} \left( \frac{R}{r} \right)^{3/4} \left( 1 - \sqrt{R/r} \right)^{1/4} \quad (18.18)$$

$$= T_{\text{disk}} \left( \frac{R}{r} \right)^{3/4} \left( 1 - \sqrt{R/r} \right)^{1/4}, \quad (18.19)$$

where

$$T_{\text{disk}} \equiv \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} \quad (18.20)$$

is a characteristic temperature of the disk. Actually,  $T_{\text{disk}}$  is roughly twice the maximum disk temperature (Problem 18.4),

$$T_{\text{max}} = 0.488 \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} = 0.488 T_{\text{disk}}, \quad (18.21)$$

which occurs at  $r = (49/36)R$ ; see Fig. 18.13.<sup>9</sup> When  $r \gg R$ , the last term on the right-hand side of Eq. (18.19) may be neglected, leaving

$$T = \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} \left( \frac{R}{r} \right)^{3/4} = T_{\text{disk}} \left( \frac{R}{r} \right)^{3/4} \quad (r \gg R). \quad (18.22)$$

This differs from our simple estimate, Eq. (18.17), by a factor of  $3^{1/4} = 1.32$ .

Integrating Eq. (18.16) for the luminosity of each ring from  $r = R$  to  $r = \infty$  results in an expression for the disk luminosity,

$$L_{\text{disk}} = G \frac{M\dot{M}}{2R}. \quad (18.23)$$

However, recall from Example 18.1.2 that without an accretion disk, the accretion luminosity (the rate at which falling matter delivers kinetic energy to the primary star) is twice as great:

$$L_{\text{acc}} = G \frac{M\dot{M}}{R}. \quad (18.24)$$

Thus, if half of the available accretion energy is radiated away as the gases spiral down through the disk, then the remaining half must be deposited at the surface of the star (or in the turbulent boundary layer between the rapidly rotating disk and the more slowly rotating primary star).<sup>10</sup>

---

**Example 18.2.1.** The maximum disk temperature,  $T_{\text{max}}$ , and the value of the disk luminosity for the white dwarf and neutron star used in Example 18.1.2 can now be evaluated. For a white dwarf with  $M = 0.85 M_{\odot}$ ,  $R = 0.0095 R_{\odot}$ , and  $\dot{M} = 10^{13} \text{ kg s}^{-1}$  ( $1.6 \times 10^{-10} M_{\odot} \text{ yr}^{-1}$ ), Eq. (18.21) is

$$T_{\text{max}} = 0.488 \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} = 2.62 \times 10^4 \text{ K}.$$

<sup>9</sup>Including the boundary layer results in  $T = 0$  where the disk meets the star's surface, an unrealistic artifact of the assumptions of the model.

<sup>10</sup>This result is just another consequence of the virial theorem.

According to Wien's displacement law, Eq. (3.19), at this temperature the blackbody spectrum peaks at a wavelength of

$$\lambda_{\max} = \frac{(500 \text{ nm})(5800 \text{ K})}{26,200 \text{ K}} = 111 \text{ nm},$$

which is in the ultraviolet region of the electromagnetic spectrum (Table 3.1). From Eq. (18.23), the luminosity of the accretion disk is

$$L_{\text{disk}} = G \frac{M\dot{M}}{2R} = 8.55 \times 10^{25} \text{ W},$$

or about  $0.22 L_{\odot}$ .

Turning now to a neutron star with  $M = 1.4 M_{\odot}$ ,  $R = 10 \text{ km}$ , and  $\dot{M} = 10^{14} \text{ kg s}^{-1}$  ( $1.6 \times 10^{-9} M_{\odot} \text{ yr}^{-1}$ ), the maximum disk temperature is

$$T_{\max} = 0.488 \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} = 6.86 \times 10^6 \text{ K}.$$

Its blackbody spectrum peaks at a wavelength of

$$\lambda_{\max} = \frac{(500 \text{ nm})(5800 \text{ K})}{686,000 \text{ K}} = 0.423 \text{ nm},$$

which is in the X-ray region of the electromagnetic spectrum. The luminosity of the neutron star's accretion disk is

$$L_{\text{disk}} = G \frac{M\dot{M}}{2R} = 9.29 \times 10^{29} \text{ W},$$

over  $2400 L_{\odot}$ . Thus the inner regions of accretion disks around white dwarfs should shine in the ultraviolet, whereas those around neutron stars will be strong X-ray sources.<sup>11</sup>

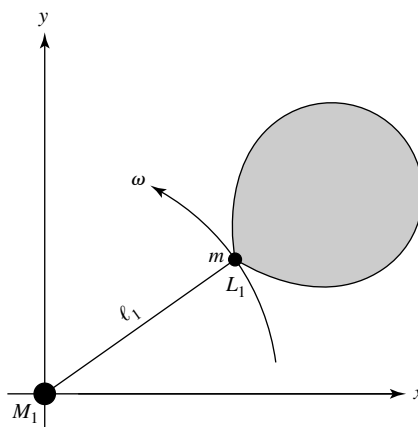
### The Radial Extent of an Accretion Disk

The radial extent of the accretion disk can be estimated by finding the value of  $r = r_{\text{circ}}$  where a continuous stream of mass that passes through  $L_1$  will settle into a circular orbit around the primary star. This may be done by considering the angular momentum of a parcel of mass  $m$  about the primary star; see Fig. 18.8. Assuming that the motion of the mass at the inner Lagrangian point is due solely to the orbital motion of the binary system, the angular momentum,  $L$ , of the mass located there is

$$L = m\omega\ell_1^2 = m\ell_1^2 \sqrt{\frac{G(M_1 + M_2)}{a^3}},$$

where Eq. (18.7) has been used for the angular frequency of the orbit, and  $\ell_1$  is given by Eq. (18.9).

<sup>11</sup>Actually, as we will see in Section 18.6, the accretion disk around a white dwarf or neutron star may be disrupted by the star's magnetic field and so may not extend down to its surface. Such systems are strong sources of X-rays.



**FIGURE 18.8** A parcel of mass  $m$  passing through the inner Lagrangian point  $L_1$ , seen from a frame of reference with the primary star at rest at the origin.

The mass  $m$  does not immediately enter into a circular orbit. Instead, the stream of mass to which  $m$  belongs flows around the primary star and collides with itself after one orbit. The orbits of the mass parcels are made circular around the primary by the collisions as energy is lost while angular momentum is conserved. When the parcel of mass has settled into a circular orbit of radius  $r_{\text{circ}}$  around  $M_1$ , its angular momentum is

$$L = m\sqrt{GM_1 r_{\text{circ}}},$$

where Eq. (2.30) was used for a circular orbit with  $\mu = mM_1/(m + M_1) \simeq m$ . Equating these two expressions for the angular momentum results in

$$\begin{aligned} r_{\text{circ}} &= a \left( \frac{\ell_1}{a} \right)^4 \left( 1 + \frac{M_2}{M_1} \right) \\ &= a \left[ 0.500 - 0.227 \log_{10} \left( \frac{M_2}{M_1} \right) \right]^4 \left( 1 + \frac{M_2}{M_1} \right). \end{aligned} \quad (18.25)$$

Since the total angular momentum must be conserved when only internal and central forces act, you may wonder what happens to the angular momentum lost by the infalling material as it spirals through the accretion disk. As shown in Pringle (1981), orbiting material that is initially in the form of a narrow ring at  $r = r_{\text{circ}}$  will spread, moving both inward and outward. The time for this migration of the disk material probably ranges from a few days to a few weeks. While most of the matter spirals inward, a small amount of the mass carries the “missing” angular momentum to the outer edge of the disk. From there, the angular momentum may be carried away from the system by wind-driven mass loss. If the accretion disk extends 80% to 90% of the way out to the inner Lagrangian point, angular momentum may also be returned to the orbital motion of the two stars by tides raised in the

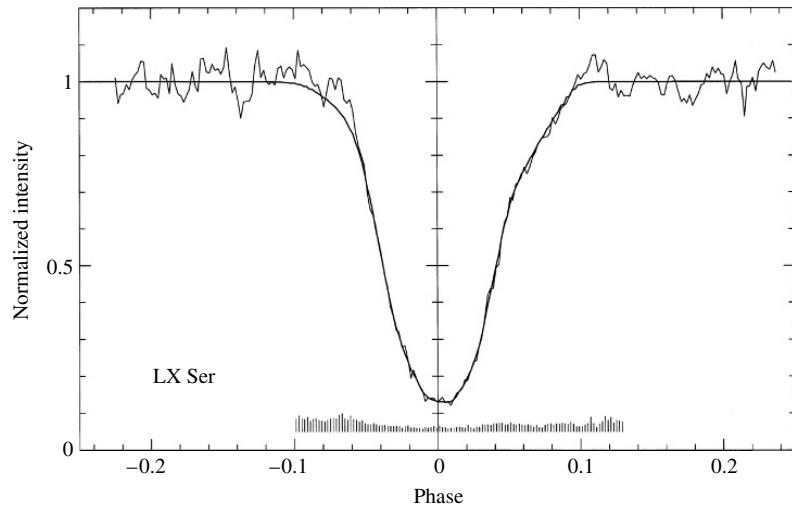
disk by the secondary star. Because of this outward migration of mass, we will adopt

$$R_{\text{disk}} \approx 2r_{\text{circ}} \quad (18.26)$$

as a rough estimate of the outer radius of the accretion disk.

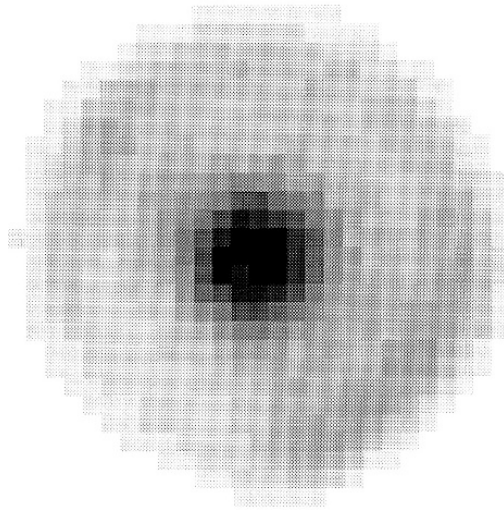
### Eclipsing, Semidetached Binary Systems

It is comforting to know that there is evidence, obtained from observing *eclipsing* semidetached binary systems, that the objects described above actually exist. Observations of light curves for eclipsing semidetached binaries, such as shown in Fig. 18.9, indicate the presence of a hot spot where the mass transfer stream collides with the outer edge of the accretion disk. The light curve can be interpreted as the result of observing consecutive “slices” of the disk as they disappear and then reappear from behind the primary star. In fact, Fig. 18.9 can be used to re-create an image of the disk itself, shown in Fig. 18.10.<sup>12</sup> Because the hot spot is on the trailing side of the disk during the eclipse (see Fig. 18.12), more light is received from the disk near the beginning of the eclipse (when the hot spot is still visible) than near the end (when the hot spot is still hidden). This produces the deficit in intensity on the right-hand side of the light curve in Fig. 18.9.



**FIGURE 18.9** The light curve of the eclipse of the accretion disk in the LX Serpentis binary system. The jagged line is the observed light curve, and the smooth line shows the fit calculated from a reconstructed image of the accretion disk, shown in Fig. 18.10. (Figure adapted from Rutten, van Paradijs, and Tinbergen, *Astron. Astrophys.*, 260, 213, 1992.)

<sup>12</sup>Using slices of the emerging disk to reconstruct an image of the accretion disk is somewhat analogous to using a CAT scan (computerized axial tomography) in a hospital to mathematically reassemble X-ray slices of the human body. Because there is more than one model disk that will reproduce a given light curve, a technique called *maximum entropy* is used to choose the smoothest possible model for the final disk image.



**FIGURE 18.10** A reconstructed negative image of the accretion disk in the LX Serpentis binary system. The hot spot on the edge of the lower right portion of the disk is smeared out in the azimuthal direction and has the appearance of a partial ring. (Figure from Rutten, van Paradijs, and Tinbergen, *Astron. Astrophys.*, 260, 213, 1992.)

### 18.3 ■ A SURVEY OF INTERACTING BINARY SYSTEMS

The life history of a close binary system is quite complicated, with many possible variations depending on the initial masses and separation of the two stars involved. As mass passes from one star to the other, the mass ratio  $M_2/M_1$  will change. The resulting redistribution of angular momentum affects the orbital period of the system as well as the separation of the two stars. The extent of the Roche lobes, given by Eqs. (18.9) and (18.10), depends on both the separation and the mass ratio of the stars, so it too will vary accordingly.

#### The Effects of Mass Transfer

The effects of mass transfer can be illustrated by considering the total angular momentum of the system. The contribution of the stars' rotation to the total angular momentum is small and may be neglected. The orbital angular momentum is given by Eq. (2.30) with an eccentricity of  $e = 0$  for a circular orbit,

$$L = \mu \sqrt{GMa}.$$

In this expression,  $\mu$  is the reduced mass (Eq. 2.22),

$$\mu = \frac{M_1 M_2}{M_1 + M_2},$$

and  $M = M_1 + M_2$  is the total mass of the two stars. Assuming (to a first approximation) that no mass or angular momentum is removed from the system via stellar winds or gravitational

radiation, both the total mass and the angular momentum of the system remain constant as mass is transferred between the two stars.<sup>13</sup> That is,  $dM/dt = 0$  and  $dL/dt = 0$ .

Some useful insights concerning the effect of the transfer of mass on the separation of the two stars can be gained by taking a time derivative of the expression for the angular momentum:

$$\begin{aligned}\frac{dL}{dt} &= \frac{d}{dt} (\mu \sqrt{GMa}) \\ 0 &= \sqrt{GM} \left( \frac{d\mu}{dt} \sqrt{a} + \frac{\mu}{2\sqrt{a}} \frac{da}{dt} \right) \\ \frac{1}{a} \frac{da}{dt} &= -\frac{2}{\mu} \frac{d\mu}{dt}.\end{aligned}\tag{18.27}$$

Remembering that the total mass,  $M$ , remains constant, we find that the time derivative of the reduced mass is

$$\frac{d\mu}{dt} = \frac{1}{M} \left( \frac{dM_1}{dt} M_2 + M_1 \frac{dM_2}{dt} \right).$$

The mass lost by one star is gained by the other. Writing  $\dot{M} \equiv dM/dt$ , this means that  $\dot{M}_1 = -\dot{M}_2$ , and so

$$\frac{d\mu}{dt} = \frac{\dot{M}_1}{M} (M_2 - M_1).$$

Inserting this into Eq. (18.27) achieves our result,

$$\frac{1}{a} \frac{da}{dt} = 2\dot{M}_1 \frac{M_1 - M_2}{M_1 M_2}.\tag{18.28}$$

Equation (18.28) describes the consequence of mass transfer on the separation of the binary system. The angular frequency of the orbit will also be affected, as shown by using Kepler's third law in the form of Eq. (18.7). Since  $M_1 + M_2 = \text{constant}$ , Kepler's third law states that  $\omega \propto a^{-3/2}$  so that

$$\frac{1}{\omega} \frac{d\omega}{dt} = -\frac{3}{2} \frac{1}{a} \frac{da}{dt}.\tag{18.29}$$

As the orbital separation decreases, the angular frequency increases.

### The Evolution of a Binary System

The following description illustrates the probable evolution of a binary system that is destined to become a cataclysmic variable. The starting point is a widely separated binary system with main-sequence stars having an initial orbital period ranging from a few months

<sup>13</sup>In fact, gravitational radiation, which will be discussed in Section 18.6, is primarily responsible for the loss of angular momentum in some short-period binary systems ( $P < 14$  hours).

to a few years. At the start, suppose that Star 1 is more massive than Star 2, so  $M_1 - M_2 > 0$ . Star 1 therefore evolves more rapidly and, depending on its mass, may become a red giant or supergiant before it begins to overflow its Roche lobe. This initiates the transfer of mass from Star 1 to Star 2 (so  $\dot{M}_1 < 0$ ). According to Eqs. (18.28) and (18.29), in this situation  $da/dt$  is negative and  $d\omega/dt$  is positive; the stars spiral closer together with an increasingly shorter period.

Now, from Eq. (18.9), as  $a$  decreases and  $M_2/M_1$  increases, the Roche lobe around Star 1 shrinks, as measured by the distance of Star 1 from the inner Lagrangian point. The mass transfer rate accelerates under the positive feedback of a shrinking Roche lobe, eventually producing an extended atmosphere around both stars, as shown in Fig. 18.4(c). The system is now a contact binary, with the degenerate core of Star 1 and the main-sequence Star 2 sharing a common gaseous envelope. The two stars transfer angular momentum to this envelope as they slowly spiral inward to a much smaller separation and shorter period. If the cores of the two stars merge, the result will be single star, which may explain the observations of blue stragglers in stellar clusters, previously discussed in Section 13.3. Alternatively, the envelope surrounding the stars may be ejected. In fact, several systems have been observed in which a binary is found at the center of a planetary nebula, possibly the result of the ejection of a common envelope. (For the sake of the following discussion, we will consider the situation where envelope ejection occurs.)

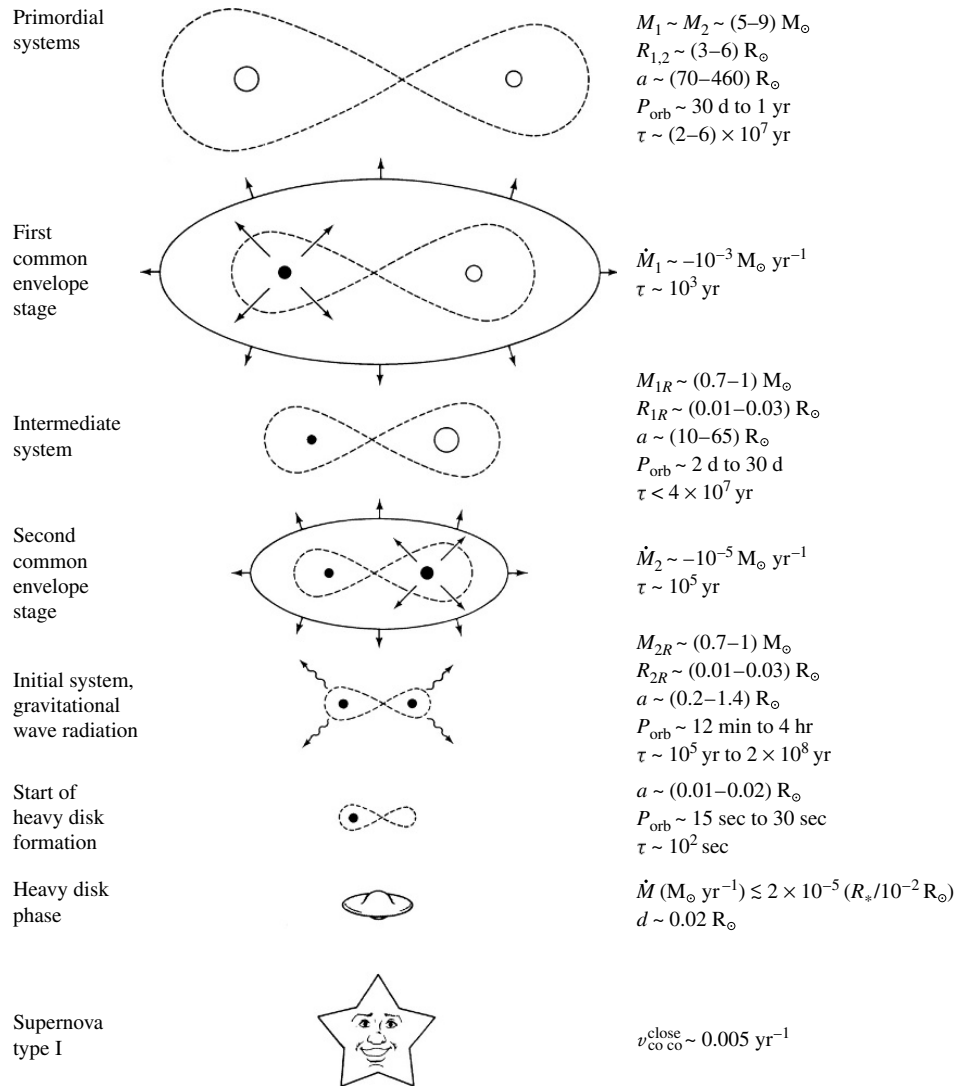
After emerging from their gaseous cocoon, the system is a detached binary; Star 2 (the secondary) lies inside its Roche lobe as Star 1 (the primary) cools to become a white dwarf. Eventually, the originally less-massive secondary star evolves and fills its Roche lobe, and mass begins to flow in the opposite direction, with  $\dot{M}_1 > 0$ . In this case a negative feedback mitigates the mass transfer process, because, as Eq. (18.28) implies, the stars will now spiral farther apart (assuming that  $M_1$  is still greater than  $M_2$ ) as the Roche lobe around the secondary star expands according to Eq. (18.10). If the mass flow is to persist, either the secondary must expand faster than the Roche lobe grows or the stars must move closer together as angular momentum is removed from the system, either by torques due to stellar winds confined by magnetic fields or by gravitational radiation.<sup>14</sup> Whatever the mechanism, a steady rate of mass transfer from the secondary to the white dwarf is maintained, and the stage is set for the outbursts of a cataclysmic variable, as will be described in Section 18.4.

As the secondary star continues to evolve, another common envelope stage may occur. Figure 18.11 shows an example of the life history of a close binary system that begins with two intermediate-mass stars (between 5 and 9  $M_\odot$ ) and culminates with two carbon–oxygen white dwarfs in a very tight orbit, circling each other every 15 s to 30 s. The larger, less massive white dwarf (recall Eq. 16.14) overflows its Roche lobe and dissolves into a heavy disk that is accreted by the more massive dwarf. The accumulation of mass pushes the primary white dwarf toward the Chandrasekhar limit, and it explodes as a Type Ia supernova.<sup>15</sup>

<sup>14</sup>The reduction of the Sun's angular momentum by the solar wind was described in Section 11.2.

<sup>15</sup>In Section 18.5, we will find that nuclear reactions begin in the core of the white dwarf before the Chandrasekhar limit is reached.





**FIGURE 18.11** One possibility for the evolution of a close binary system, ending in a Type Ia supernova. The masses and radii of the stars, and their orbital separation ( $A$ ), orbital period, and mass transfer rate, are given for some stages, along with the duration ( $\tau$ ). (Figure adapted from Iben and Tutukov, *Ap. J. Suppl.*, 54, 335, 1984.)

### Types of Interacting Binary Systems

There are many types of close binary systems, too many to discuss in any detail. The following list<sup>16</sup> describes the main classes of interacting binaries, together with some of the features that make these systems important for astronomers. Many of the classes are named after the prototype object for that class.

- **Algols.** These are two normal stars (main-sequence stars or subgiants) in a semidetached binary system. They provide checks on stellar properties and evolution, and they yield information on mass loss and mass exchange. Active Algols (the W Serpens stars) provide laboratories for studying rapid (short-lived) stages of stellar and binary star evolution. These systems are important for studying accretion processes and accretion disks. Mass loss from Algols may contribute to the chemical enrichment of the interstellar medium.
- **RS Canum Venaticorum and BY Draconis Stars.** These stars are chromospherically active binaries that are important systems for investigating dynamo-driven magnetic activity in cool stars (spectral type F and later). Manifestations of enhanced magnetic activity include starspots, chromospheres, coronae, and flares. These systems also contribute to our understanding of the magnetic activity of the Sun—the so-called *solar–stellar connection*.
- **W Ursae Majoris Contact Systems.** These short-period (0.2–0.8 day) contact binaries display very high levels of magnetic activity and are important stars for studying the stellar dynamo mechanism at extreme levels. The drag of magnetic braking may cause these binaries to coalesce into single stars.
- **Cataclysmic Variables and Nova-like Binaries.** These systems have short periods and contain white dwarf components together with cool M-type secondaries that fill their Roche lobes. They provide valuable information on the final stages of stellar evolution. These binaries are also important for studying accretion phenomena and accretion disk properties.
- **X-ray Binaries with Neutron Star and Black Hole Components.** These systems are powerful ( $L_x > 10^{28}$  W) X-ray sources that have neutron star or (more rarely) black hole components. The X-rays are due to the accretion of gas onto the degenerate component of the system from a nondegenerate companion. Observations of neutron star systems supplement the information on their structure and evolution that comes from pulsars (such as masses, radii, rotation, and magnetic fields). Systems such as V404 Cygni, A0620–00, and Cygnus X-1 provide evidence for the existence of black holes; see Section 18.6.
- **ζ Aurigae and VV Cephei Systems.** These long-period interacting binaries contain a late-type supergiant component and a hot (usually spectral type B) companion. ζ-Aur

<sup>16</sup>Quoted with permission from E. F. Guinan, *Evolutionary Processes in Interacting Binary Stars*, Kondo, Sisteró, and Polidan (eds.), Kluwer Academic Publishers, Dordrecht, 1992. Reprinted by permission of Kluwer Academic Publishers.

systems contain G or K supergiant stars, and VV Cep binaries contain M supergiants. Although not originally interacting binaries, they became so when the more massive star evolved to become a supergiant. When eclipses occur, the atmosphere and wind of the cooler supergiant can be probed as the hotter star passes behind.

- **Symbiotic Binaries.** Symbiotic stars are long-period interacting binaries consisting of an M giant (sometimes a pulsating Mira-type variable) and an accreting component that can be a white dwarf, subdwarf, or low-mass main-sequence star. The common feature of these systems is the accretion of the cool component's wind onto its hot companion. Orbital periods of symbiotic stars typically range between 200 and 1500 days. Several of the symbiotic binaries have the cool component filling its Roche lobe, making them symbiotic Algol systems.
- **Barium and S-Star Binaries.** These stars are thought to be long-period binaries in which the originally more massive component evolved and transferred some of its nuclear-processed gas to the present K or M giant companion (recall the discussion of spectral class S on page 466). The giant stars are thought to have white dwarf companions that are often too cool to be seen in the ultraviolet. These systems are important for studying nucleosynthesis and mass loss in evolved stars.
- **Post-Common-Envelope Binaries.** These binary systems usually contain hot white dwarf or subdwarf components and cooler secondary stars that have presumably passed through the common envelope phase of binary star evolution. The binary nuclei of planetary nebulae are examples of post-common-envelope binaries. These systems are important for studying short-lived stages of stellar evolution.

## 18.4 ■ WHITE DWARFS IN SEMIDETACHED BINARIES

When a white dwarf is the primary component of a semidetached binary system, the result may be a **dwarf nova**, a **classical nova**, or a **supernova**, in order of increasing brilliance. It is somewhat unfortunate that the term *nova* (Latin for “new”) appears in each name, because the three types of outbursts employ three very different mechanisms.

### Cataclysmic Variables

Dwarf novae and classical novae belong to the general class of **cataclysmic variables**, of which more than one thousand systems are known to exist. They survive their release of energy (unlike supernovae), and the outburst process can reoccur. Cataclysmic variables are characterized by long quiescent intervals punctuated by outbursts in which the brightness of the system increases by a factor between 10 (for dwarf novae) and  $10^6$  (for classical novae). The mean mass of the primary star is  $0.86 M_{\odot}$ , which is larger than the average of about  $0.58 M_{\odot}$  for isolated white dwarfs. The secondary star is usually a main-sequence star of spectral type G or later and is less massive than the primary star.

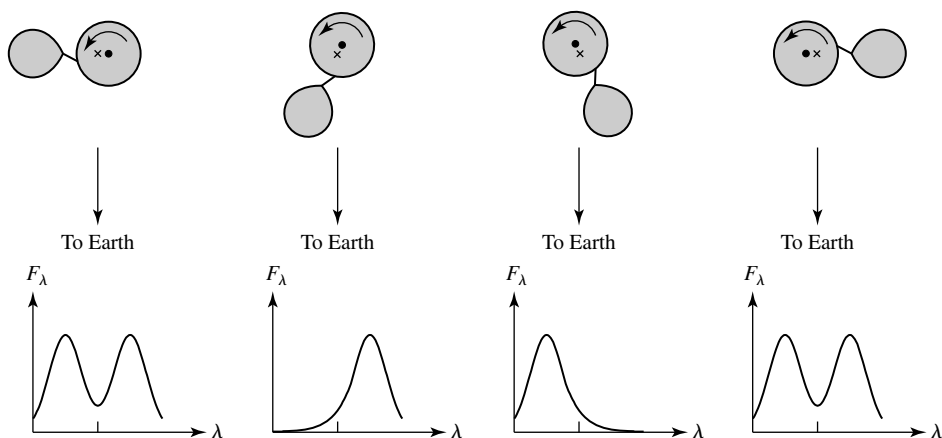
The two stars orbit each other with periods ranging from 23 minutes to more than five days, although the vast majority have orbital periods of between 78 minutes and 12 hours. Interestingly, a “period gap” exists in the orbital periods of cataclysmic variables between

1.5 hours and 3.25 hours; it is probably due to an abrupt change in angular momentum transfer in the system, associated with a complex interplay of disrupted magnetic braking, gravitational radiation, the changing size of Roche lobes, and the evolution of the stars.

The outbursts are believed to be due to a sudden increase in the rate at which mass flows down through the disk. As the eclipsed disk emerges from behind the secondary star, the radial variation in the disk's temperature can be determined. During an outburst, the disk does indeed appear to be optically thick, with  $T \propto r^{-3/4}$ , in agreement with Eq. (18.22). But during quiescence the observations are not consistent with the disk model described above, probably because the disk is not completely optically thick when it is cooler and contains less mass.

Additional evidence supporting this view comes from the strong, wide emission lines of hydrogen and helium that are seen in cataclysmic variables during quiescence. These lines are usually doubly peaked, as shown in Fig. 18.12. However, during an eclipse a single emission line is observed, either redshifted or blueshifted. This is what would be expected from a rotating disk of optically thin gas; the Doppler-shifted emission lines produced on the opposite sides of the disk disappear when one side or the other is hidden behind the secondary star.

The source of the emission lines that appear during a cataclysmic variable's quiescent phase is not yet clear. During an outburst, these lines appear in absorption, as would be expected from an optically thick disk that produces absorption lines in the same manner as an optically thick stellar atmosphere. But during quiescence, the rate at which mass flows down through the disk has presumably decreased, making the disk less dense and cooler. At larger radii the disk may then be optically thin and so produce emission lines. Alternatively, there may be a thin layer of hot gas above the disk that produces the emission lines.



**FIGURE 18.12** A Doppler-shifted emission line at different stages of the eclipse of an accretion disk. The binary system orbits about its center of mass (the “x”), and is observed nearly edge-on. The disk rotates in the direction indicated by the arrow.

### Dwarf Novae

The first observation of a dwarf nova (U Geminorum) was made in 1855. However, the basic nature of these objects remained elusive until 1974, when Brian Warner at the University of Cape Town showed that the outburst of an eclipsing dwarf nova, Z Chamaeleontis, was due to a brightening of the accretion disk surrounding the white dwarf. Since most of the light from a dwarf nova comes from the accretion disk around the white dwarf, these systems provide astronomers with their best opportunity to study the dynamic structure of accretion disks.<sup>17</sup> Observations of the dwarf nova VW Hydri showed that the outburst at visible wavelengths preceded the ultraviolet brightening by about a day. This indicates that the outburst started in the cooler, outer part of the disk and then spread down to the hotter central regions. For these reasons, astronomers have concluded that the outbursts of dwarf novae are caused by a sudden increase in the rate at which mass flows down through the accretion disk.

---

**Example 18.4.1.** Z Chamaeleontis is a dwarf nova. It consists of an  $M_1 = 0.85 M_\odot$  white dwarf primary with a radius of  $R = 0.0095 R_\odot$  and a late M-type main-sequence secondary star of mass  $M_2 = 0.17 M_\odot$ . The orbital period of the system is  $P = 0.0745$  day. What does this system look like?

From Kepler's third law, Eq. (2.37), the separation of the two stars is

$$a = \left[ \frac{P^2 G (M_1 + M_2)}{4\pi^2} \right]^{1/3} = 5.22 \times 10^8 \text{ m},$$

about 75% of the radius of the Sun. The distance between the white dwarf primary and the inner Lagrangian point  $L_1$  is given by Eq. (18.9),

$$\ell_1 = a \left[ 0.500 - 0.227 \log_{10} \left( \frac{M_2}{M_1} \right) \right] = 3.44 \times 10^8 \text{ m}.$$

Because the secondary star fills its Roche lobe in a semidetached binary system, the distance between the secondary star and the inner Lagrangian point is a measure of the size of the secondary. For Z Cha,

$$R_2 \approx \ell_2 = a - \ell_1 = 1.78 \times 10^8 \text{ m},$$

which agrees quite well with the size of an M6 main-sequence star (see Appendix G).

The value of  $r_{\text{circ}}$  for this system is, from Eq. (18.25),

$$r_{\text{circ}} = a \left( \frac{\ell_1}{a} \right)^4 \left( 1 + \frac{M_2}{M_1} \right) = 1.18 \times 10^8 \text{ m},$$

*continued*

<sup>17</sup>In some systems, the primary white dwarf has a magnetic field that is sufficiently strong (a few thousand teslas) to prevent the formation of an accretion disk. Instead, the accretion takes place through a magnetically controlled column that funnels mass onto one (or both) of the white dwarf's magnetic poles. These *AM Herculis stars* (or *polars*) will be considered in Section 18.6.

and so a crude estimate of the outer radius of the disk is

$$R_{\text{disk}} \approx 2r_{\text{circ}} = 2.4 \times 10^8 \text{ m},$$

(Eq. 18.26), which is about two-thirds of the way to the inner Lagrangian point. This is in good agreement with observations that indicate that the Z Cha's disk emits very little light from beyond this radius.

The mass transfer rate inferred for Z Cha during an outburst is roughly

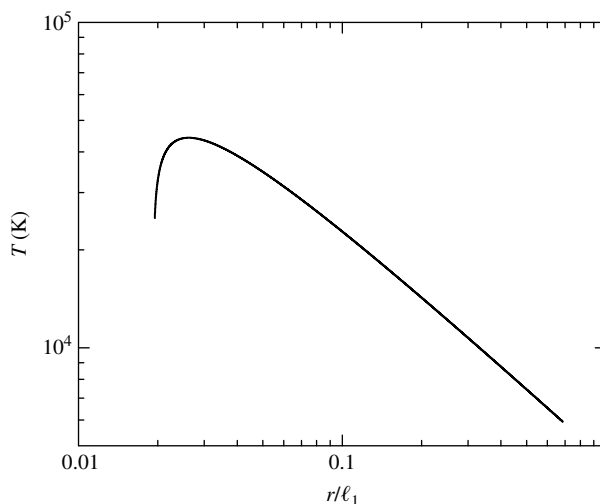
$$\dot{M} = 1.3 \times 10^{-9} M_{\odot} \text{ yr}^{-1},$$

or  $7.9 \times 10^{13} \text{ kg s}^{-1}$ , which implies a maximum disk temperature of

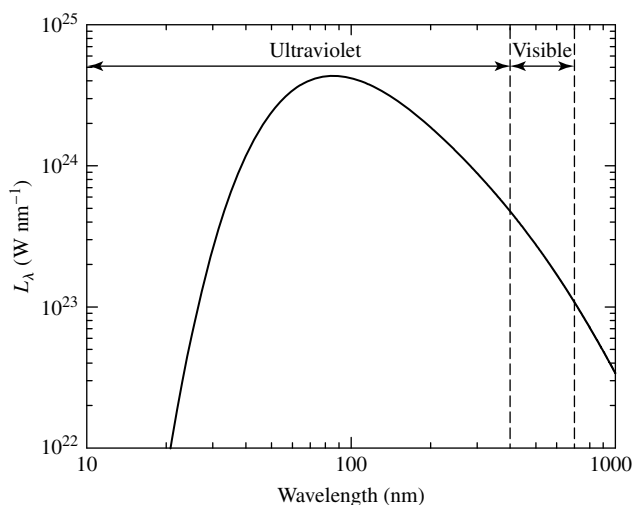
$$T_{\text{max}} = 0.488 \left( \frac{3GM\dot{M}}{8\pi\sigma R^3} \right)^{1/4} = 4.4 \times 10^4 \text{ K},$$

using Eq. (18.21). Figure 18.13 shows the variation in the disk temperature with radius for Z Cha [calculated from Eq. (18.19)]. Moving from the inner to the outer regions of the disk, the temperature falls from 44,000 K to 8000 K. According to Wien's law, Eq. (3.15), this corresponds to an increase in the peak wavelength of the emitted radiation from 66 nm to 363 nm (from the far to the near portions of the ultraviolet spectrum).

The monochromatic luminosity,  $L_{\lambda}$ , for the entire disk can be calculated by integrating Eq. (3.22) for the Planck function,  $B_{\lambda}$ , over the disk area and over all directions (recall



**FIGURE 18.13** The temperature of the accretion disk calculated for the dwarf nova Z Chamaeleontis. The radius  $r$  is given in units of  $\ell_1$ , the distance from the white dwarf to the inner Lagrangian point. The sudden drop in temperature near the surface of the white dwarf primary is an unrealistic artifact of the assumptions.



**FIGURE 18.14** The monochromatic luminosity calculated for the accretion disk of the dwarf nova Z Chamaeleontis.

Section 3.5). The resulting graph of the energy emitted per second within wavelength intervals of 1 nm is shown in Fig. 18.14. According to Eq. (18.23), the total luminosity of the accretion disk (integrated over all wavelengths) is

$$L_{\text{disk}} = G \frac{M \dot{M}}{2R} = 6.8 \times 10^{26} \text{ W},$$

which exceeds the luminosity of the Sun by about 75%.

An artist's conception of the appearance of Z Cha is shown in Fig. 18.15; see also Fig. 18.6.

### Changes in the Mass Transfer Rate

To date, more than 250 dwarf novae have been discovered. Characteristically, they brighten by between 2 and 6 magnitudes (factors of between 6 and 250 in luminosity) during outbursts that usually last from about 5 to 20 days. These eruptions are separated by quiet intervals of 30–300 days; see Fig. 18.16. Estimates of the rate of mass transfer through the disks of dwarf novae have been obtained by comparing theoretical models with observations of the amount of energy released at different wavelengths. Apparently, during the long quiescent intervals,

$$\dot{M} \approx 10^{12} - 10^{13} \text{ kg s}^{-1} \approx 10^{-11} - 10^{-10} M_{\odot} \text{ yr}^{-1},$$

which increases to

$$\dot{M} \approx 10^{14} - 10^{15} \text{ kg s}^{-1} \approx 10^{-9} - 10^{-8} M_{\odot} \text{ yr}^{-1}$$



**FIGURE 18.15** An artist's conception of Z Chamaeleontis. [Courtesy of Dale W. Bryner (1935–1999), Weber State University.]

during an outburst. Since the disk luminosity is proportional to  $\dot{M}$  (Eq. 18.23), this increase in the mass transfer rate by a factor of 10–100 is consistent with the observed brightening of the system.<sup>18</sup>

The mystery remaining to be solved by astronomers is the origin of the increased rate of mass transfer through the disk of a dwarf nova during an outburst. Possible explanations focus on either an instability in the mass transfer rate from the secondary to the primary star or an instability in the accretion disk itself that periodically dams up and releases the gases flowing through it.

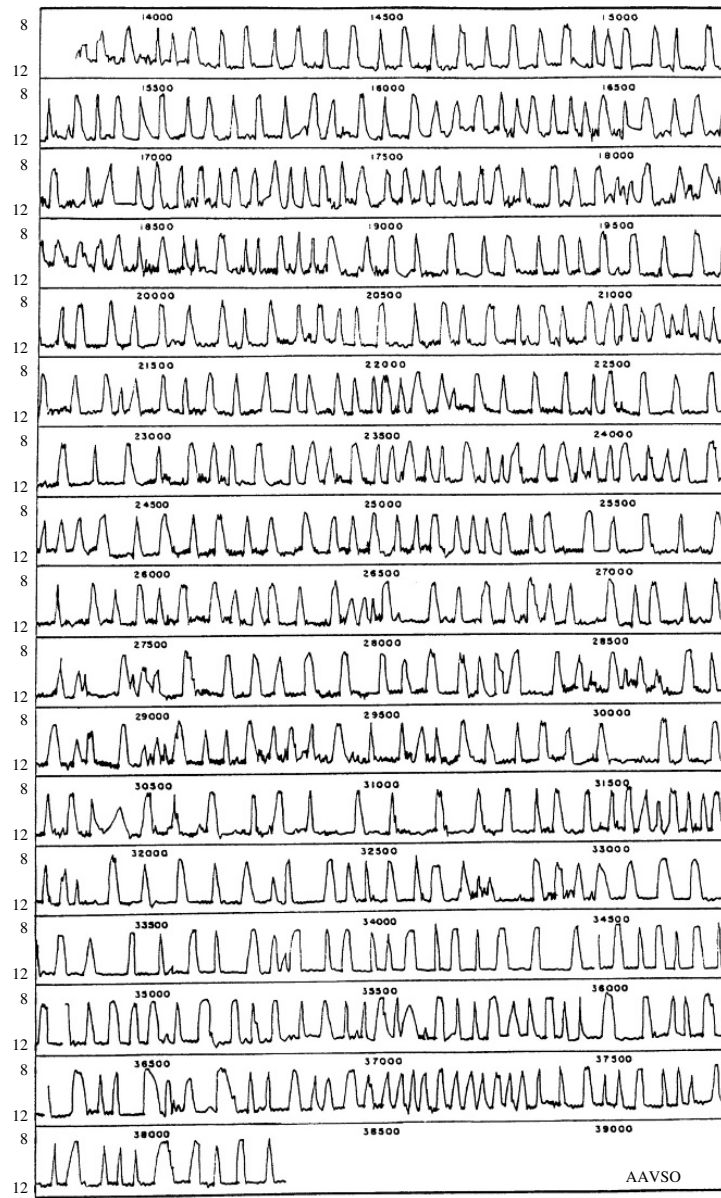
A modulation of the mass transfer rate must depend on the details of the mass flow through the inner Lagrangian point,  $L_1$ . One possibility is an instability in the outer layers of the secondary star, causing it to periodically overflow its Roche lobe. Such an instability could be powered by the hydrogen partial ionization zone (at  $T \approx 10,000$  K) damming up and releasing energy.<sup>19</sup> When one kilogram of H II ions recombines with free electrons, as much as  $1.3 \times 10^9$  J is released. If the ionization zone occurs close enough to the surface of the secondary, this could be sufficient to propel some of the overlying stellar material through the  $L_1$  point and initiate a dwarf nova outburst. Recall, however, that the secondary star is usually a main-sequence star of spectral type G or later, so the ionization zone may well lie too deep to produce the instability.

The alternative explanation involving an instability in the outer part of the accretion disk also utilizes the hydrogen partial ionization zone. The viscosity of the disk material governs the rate at which mass spirals down through the disk. The lower the viscosity, the lower the resistance to the orbital motion of the disk gases; the inward drift of material decreases, and more matter accumulates in the disk. If the viscosity periodically switches

<sup>18</sup>Recall that a difference of 5 magnitudes corresponds to a factor of 100 in brightness; see Eq. (3.4).

<sup>19</sup>This is somewhat reminiscent of the  $\kappa$ -mechanism that is involved in stellar pulsation; recall Section 14.2.





Light curve of SS Cygni  
1896–1963

**FIGURE 18.16** Outbursts of the dwarf nova SS Cygni, about 95 pc away. This light curve, labeled by Julian day at 500-day intervals, covers the years 1896–1963 and was compiled by the American Association of Variable Star Observers. (We acknowledge with thanks the variable star observations from the AAVSO International Database contributed by observers worldwide.)

from a low to a high value, the resulting wave of stored material plunging inward could produce the brightening of the disk observed for dwarf novae. Although the source of the viscosity in accretion disks is poorly understood, it has been suggested that the switch between low and high viscosity may be produced by an instability involving the periodic ionization and recombination of hydrogen in the outer part of the disk where  $T \approx 10,000$  K. In such a scenario, the viscosity is roughly proportional to the disk temperature, which in turn depends on the opacity of the disk material. Below  $10^4$  K, a plausible chain of reasoning then suggests

neutral hydrogen  $\rightarrow$  low opacity  
 $\rightarrow$  efficient cooling  
 $\rightarrow$  low temperature  
 $\rightarrow$  low viscosity  
 $\rightarrow$  mass retained in the outer disk.

On the other hand, above  $10^4$  K,

ionized hydrogen  $\rightarrow$  high opacity  
 $\rightarrow$  inefficient cooling  
 $\rightarrow$  high temperature  
 $\rightarrow$  high viscosity  
 $\rightarrow$  mass released to fall through the disk.

The instability occurs because the accumulation of matter tends to slowly heat the outer disk, while its release results in a rapid cooling. This mechanism should operate only for low accretion rates ( $< 10^{12} \text{ kg s}^{-1} \approx 10^{-11} M_{\odot} \text{ yr}^{-1}$ ), so dwarf novae outbursts should not occur for systems with larger values of  $\dot{M}$ . This limit is in fact observed and is one reason why most astronomers favor the disk instability explanation of dwarf novae outbursts.

### Classical Novae

Higher accretion rates are associated with classical novae. The earliest record of a nova was that of CK Vulpeculae, which occurred in 1670. Since then hundreds of others have been observed. About 30 novae are detected in the Andromeda galaxy (M31) each year, but only two or three per year can be seen in those regions of our own Milky Way Galaxy that are unobscured by dust. Novae are characterized by a sudden increase in brightness of between 7 and 20 magnitudes, with an average brightening of about 10–12 magnitudes. The rise in luminosity is very rapid, taking only a few days, with a brief pause or *standstill* when the star is about two magnitudes from its maximum brilliance. At its peak, a nova may shine with about  $10^5 L_{\odot}$  and release roughly  $10^{38}$  J (integrated across all wavelengths) over  $\sim 100$  days.